Computers, brains and AI

3: Logic (silicon and computers)

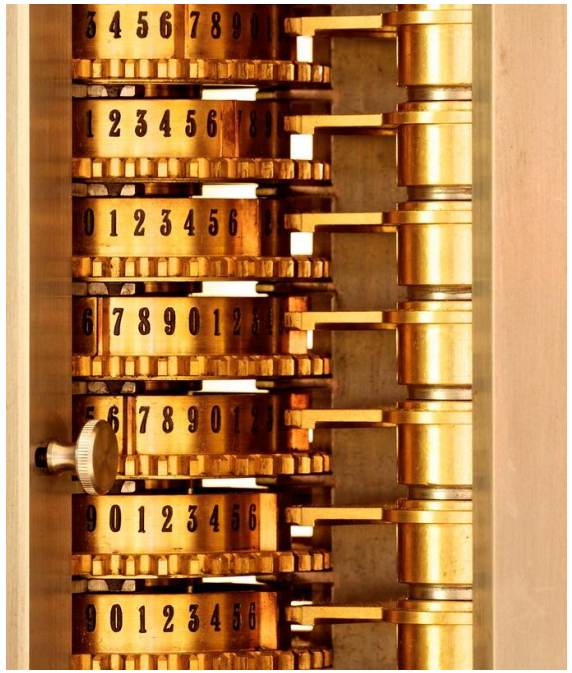




Photo by Marcin Wichary / CC BY
Photo by Science Museum Group / CC0

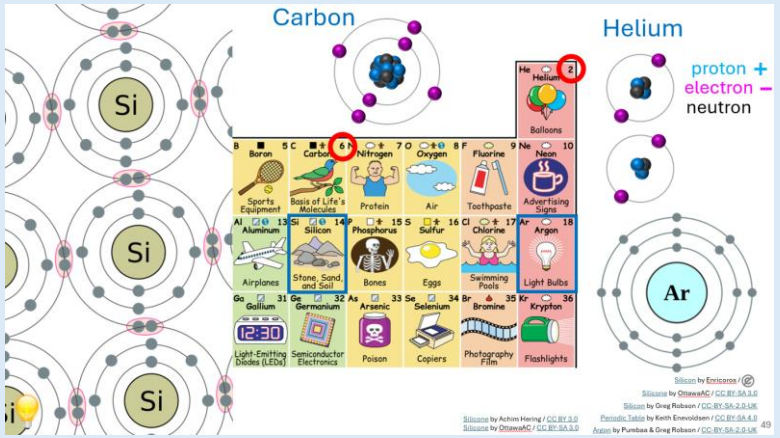
Today...

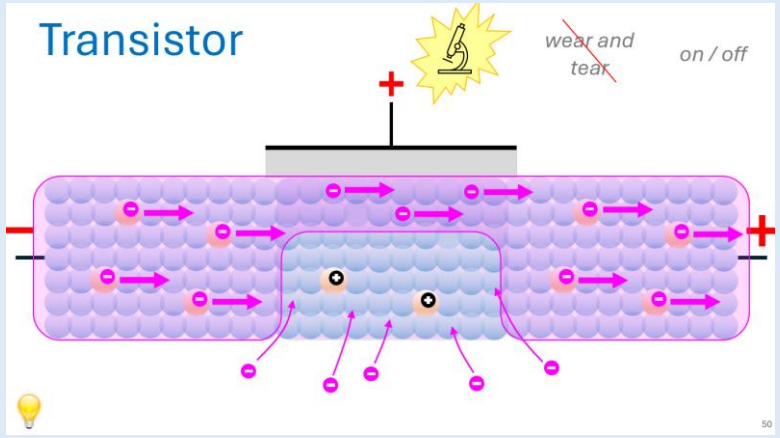
Silicon

Computers

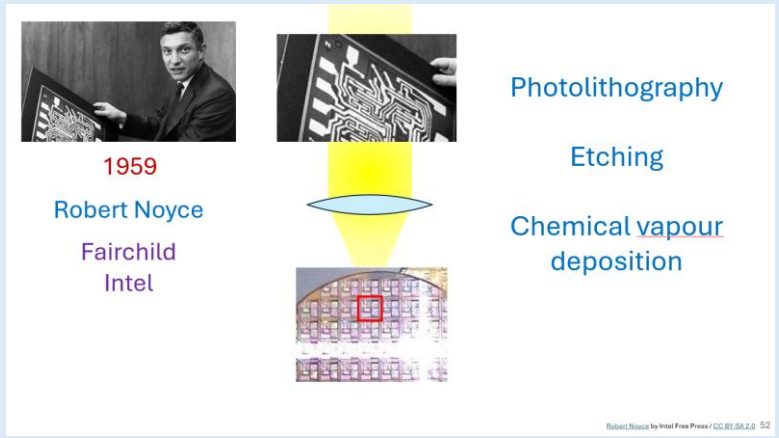
1) Intro
2) Foundations
3) ~~Logic~~ Silicon
4) Deep learning
5) Deep learning
6) ChatGPT
7) ChatGPT
8) ChatGPT
9) The future of AI
10) The future of AI

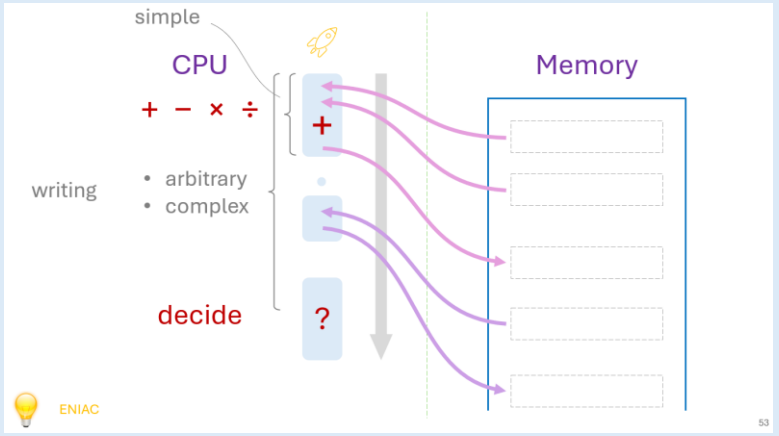


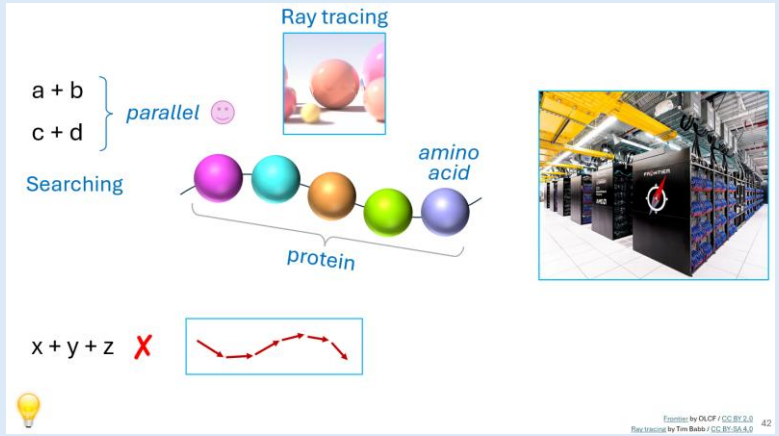

Carbon
Helium
proton +
electron −
neutron
Si
Si
Si
Ar
Balloons
Sports Equipment
Basis of Life's Molecules
Protein
Air
Toothpaste
Advertising Signs
Airplanes
Stone, Sand and Soil
Bones
Eggs
Swimming Pools
Light Bulbs
Light-Emitting Diodes (LEDs)
Semiconductor Electronics
Poison
Copiers
Photography Film
Flashlights


Transistor
wear and tear
on / off
+
+
−

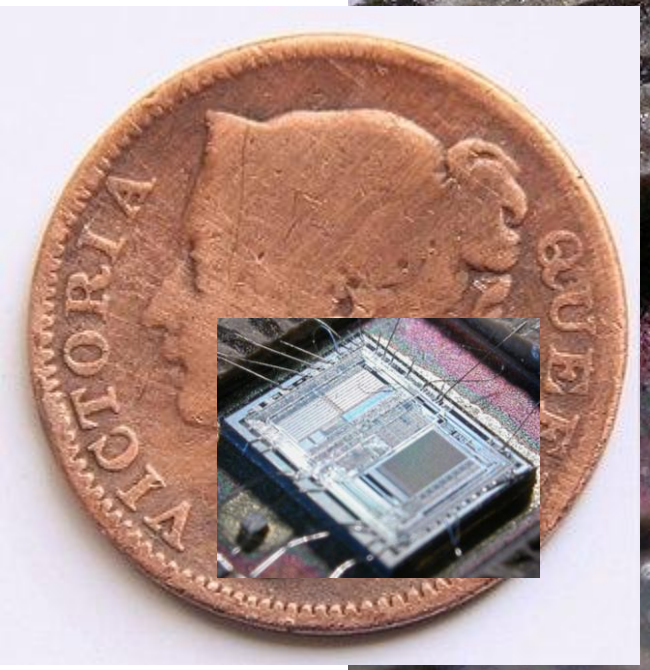

Jan Czochralski
1915
sand
silicon
99.9999999%
quartz
1956
Czochralski process
crystal
Silicon ingot


1959
Robert Noyce
Fairchild
Intel
Photolithography
Etching
Chemical vapour deposition


simple
CPU
+ − × ÷
writing
• arbitrary
• complex
decide
?
Memory
ENIAC

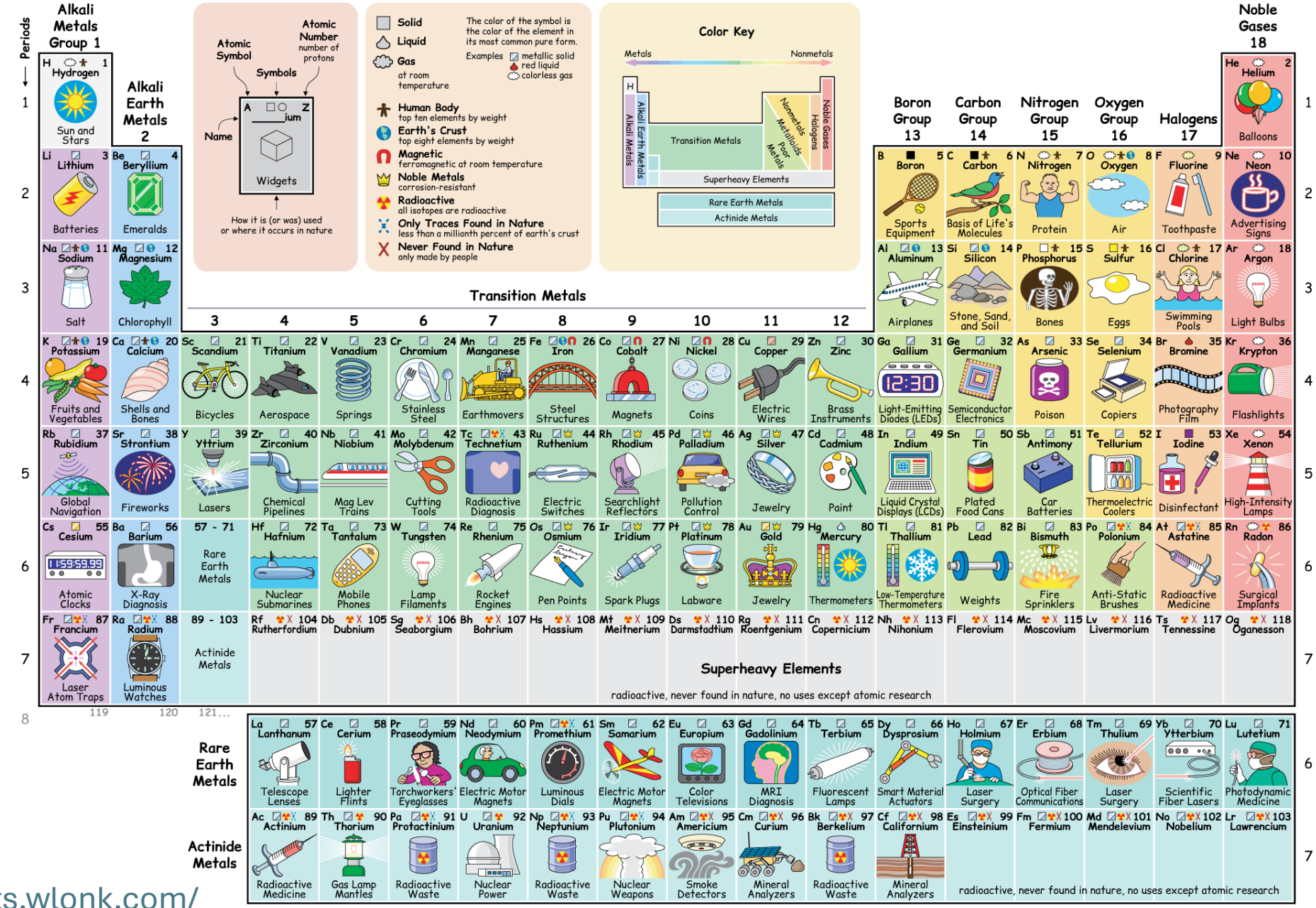

Ray tracing
a + b
c + d
parallel
Searching
amino acid
protein
x + y + z ✗

BBC Horizon – 1978
“Now the chips are down”



Coin By AKS.9955 / CC BY-SA 4.0
Integrated circuit by Ioan Sameli / CC BY-SA 2.0

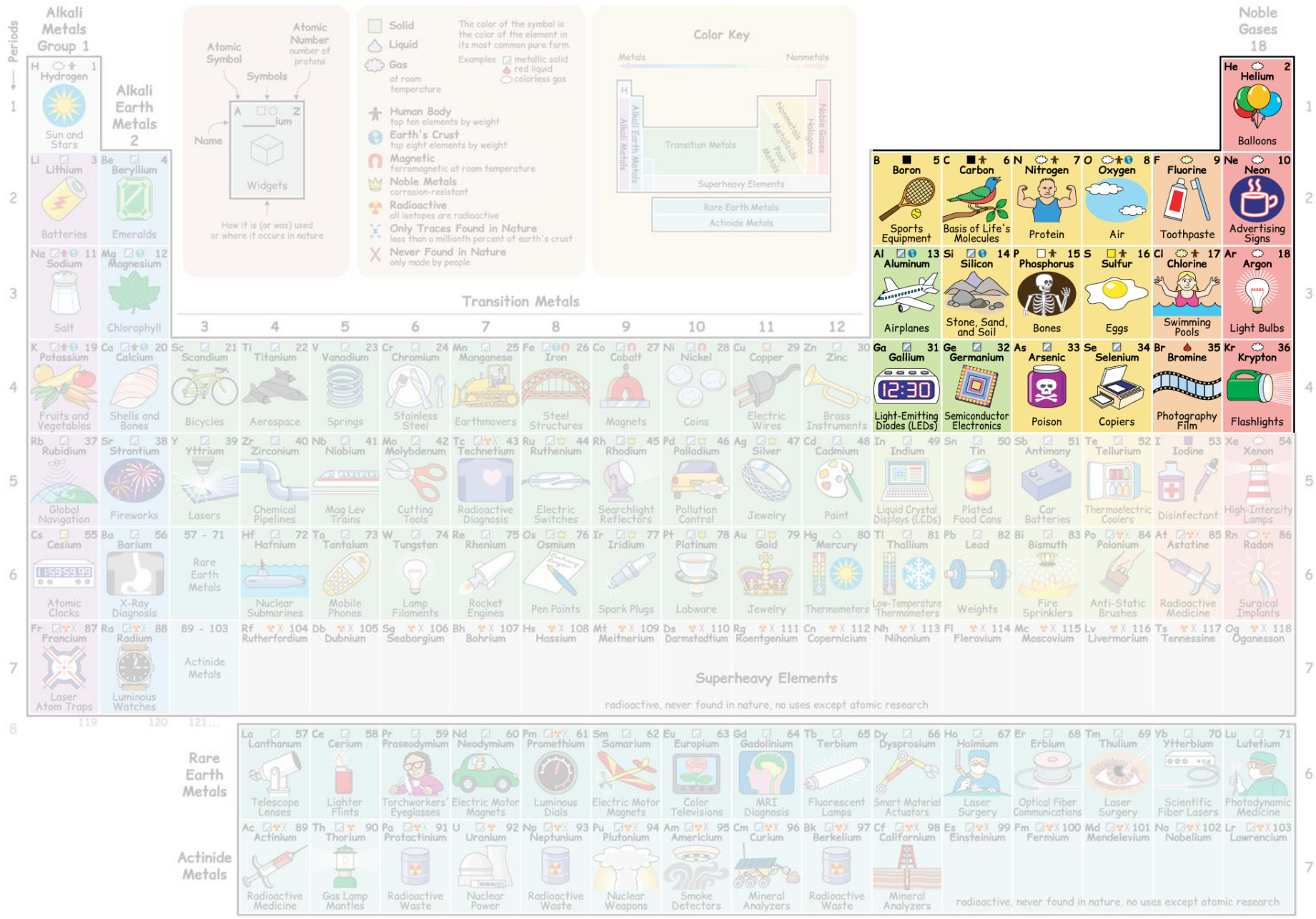
The Periodic Table of the Elements, in Pictures



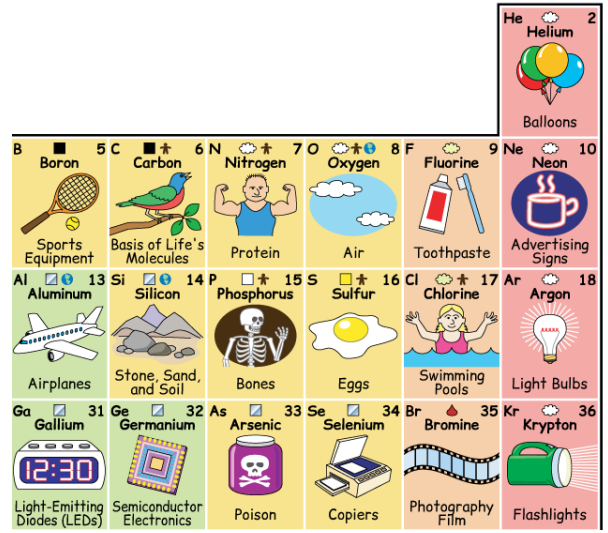
https://elements.wlonk.com/

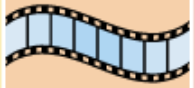
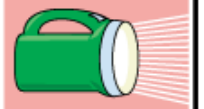
Periodic Table
by Keith Enevoldsen
CC BY-SA 4.0

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Periodic Table
by Keith Enevoldsen
CC BY-SA 4.0


He 2 Helium
Balloons
B 5 Boron
Sports Equipment
C 6 Carbon
Basis of Life's Molecules
N 7 Nitrogen
Protein
O 8 Oxygen
Air
F 9 Fluorine
Toothpaste
Ne 10 Neon
Advertising Signs
Al 13 Aluminum
Airplanes
Si 14 Silicon
Stone, Sand, and Soil
P 15 Phosphorus
Bones
S 16 Sulfur
Eggs
Cl 17 Chlorine
Swimming Pools
Ar 18 Argon
Light Bulbs
Ga 31 Gallium
12:30
Light-Emitting Diodes (LEDs)
Ge 32 Germanium
Semiconductor Electronics
As 33 Arsenic
Poison
Se 34 Selenium
Copiers
Br 35 Bromine
Photography Film
Kr 36 Krypton
Flashlights

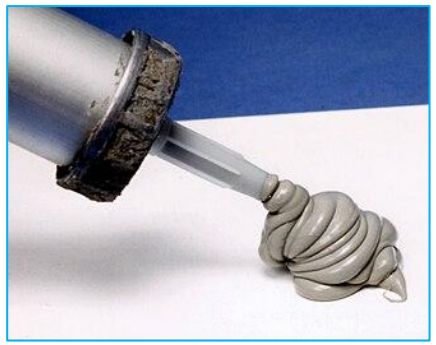
					He 2 Helium  Balloons
B 5 Boron  Sports Equipment	C 6 Carbon  Basis of Life's Molecules	N 7 Nitrogen  Protein	O 8 Oxygen  Air	F 9 Fluorine  Toothpaste	Ne 10 Neon  Advertising Signs
Al 13 Aluminum  Airplanes	Si 14 Silicon  Stone, Sand, and Soil	P 15 Phosphorus  Bones	S 16 Sulfur  Eggs	Cl 17 Chlorine  Swimming Pools	Ar 18 Argon  Light Bulbs
Ga 31 Gallium  Light-Emitting Diodes (LEDs)	Ge 32 Germanium  Semiconductor Electronics	As 33 Arsenic  Poison	Se 34 Selenium  Copiers	Br 35 Bromine  Photography Film	Kr 36 Krypton  Flashlights

Silicon

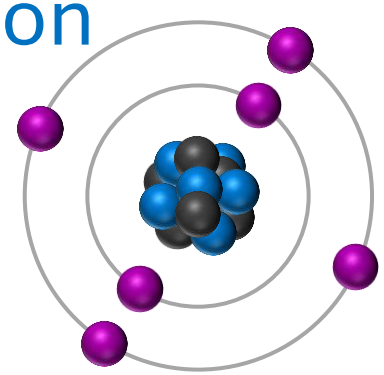


Silicone



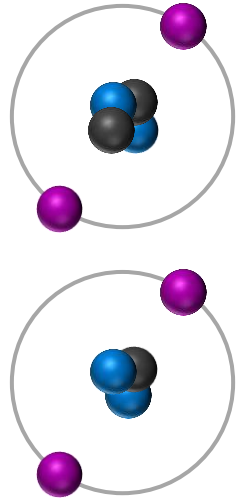


Carbon

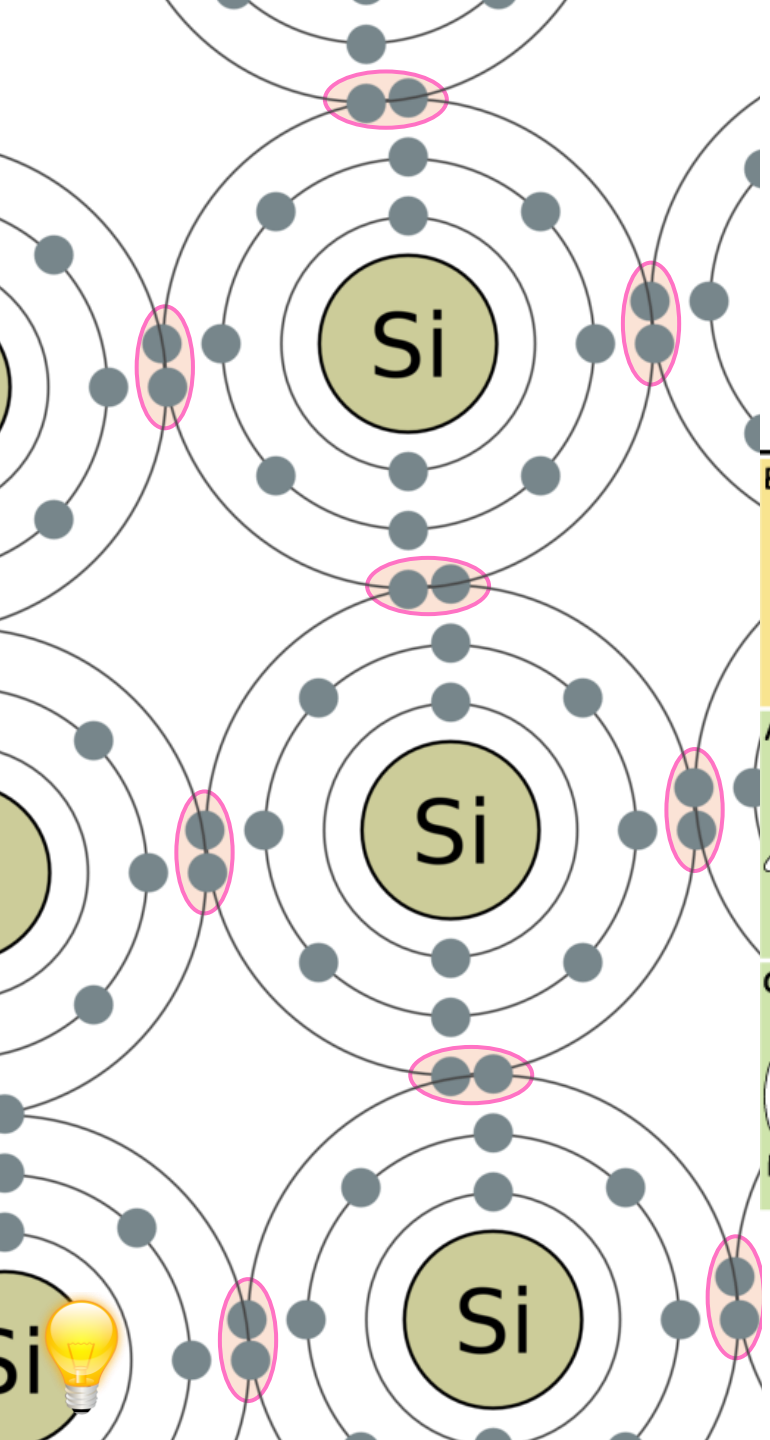




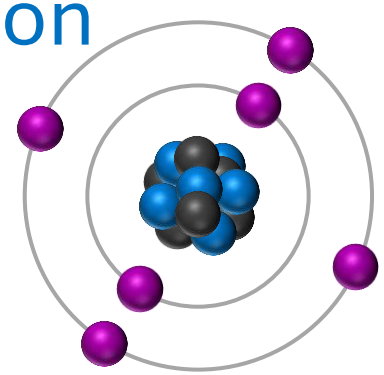
Helium

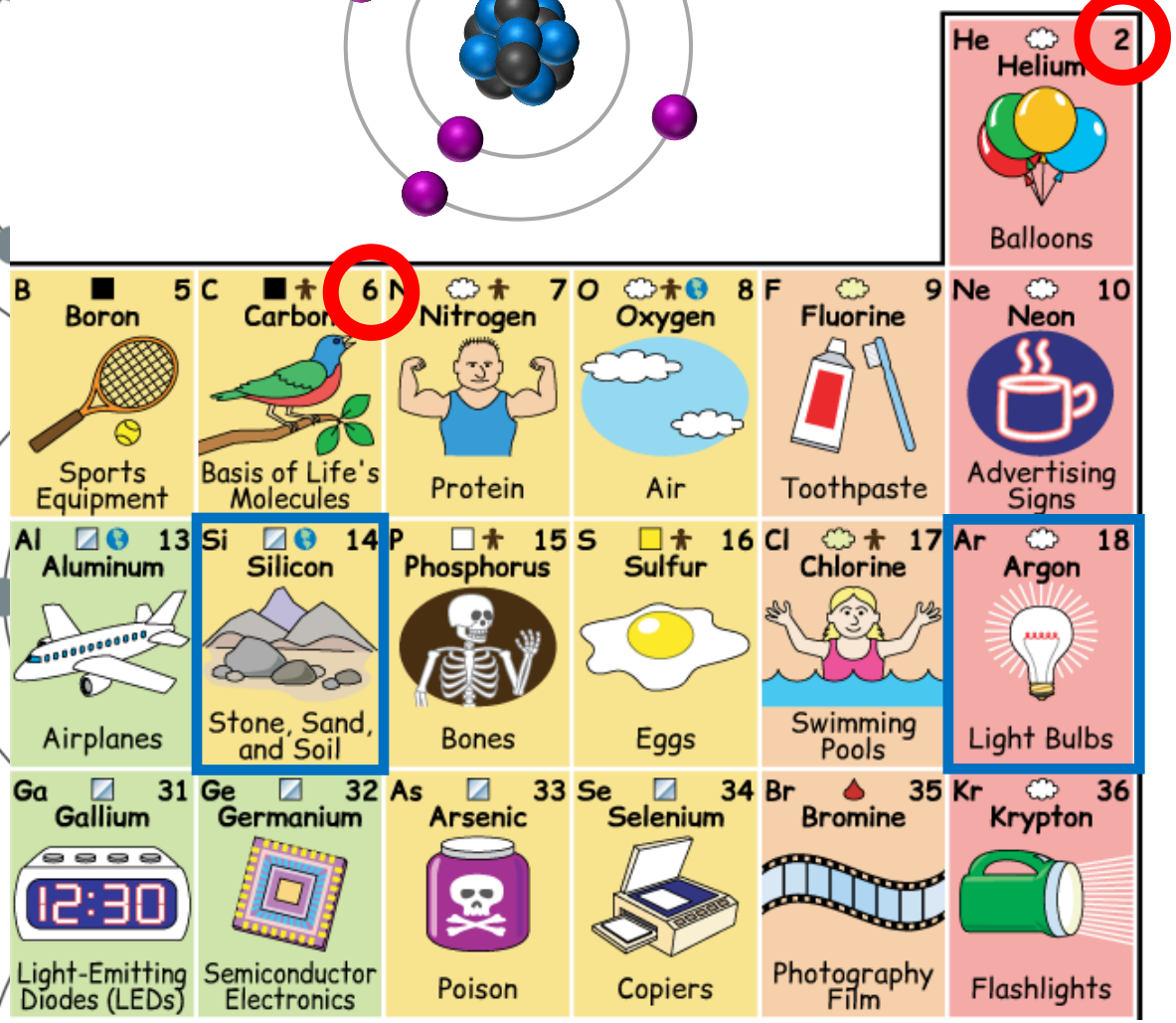


proton +
electron –
neutron

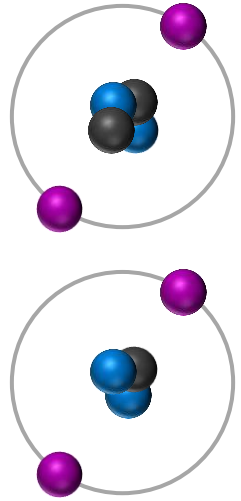


Carbon

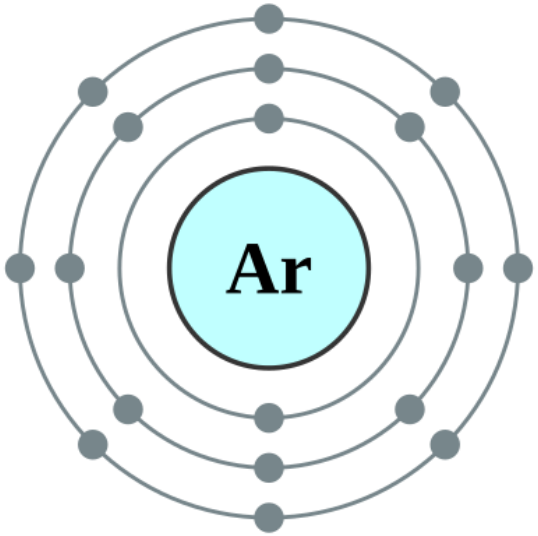




Helium

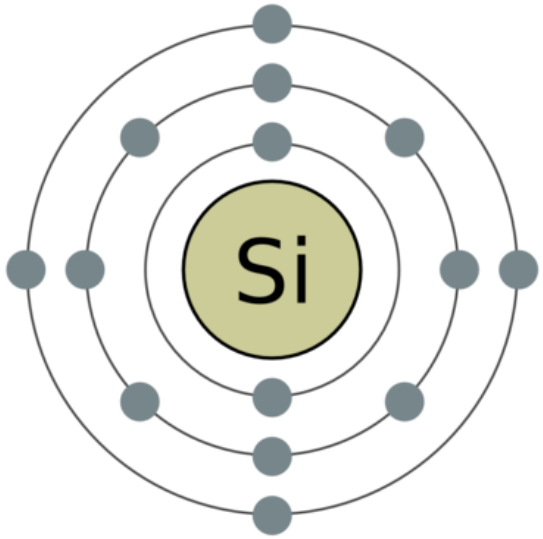


proton +
electron –
neutron



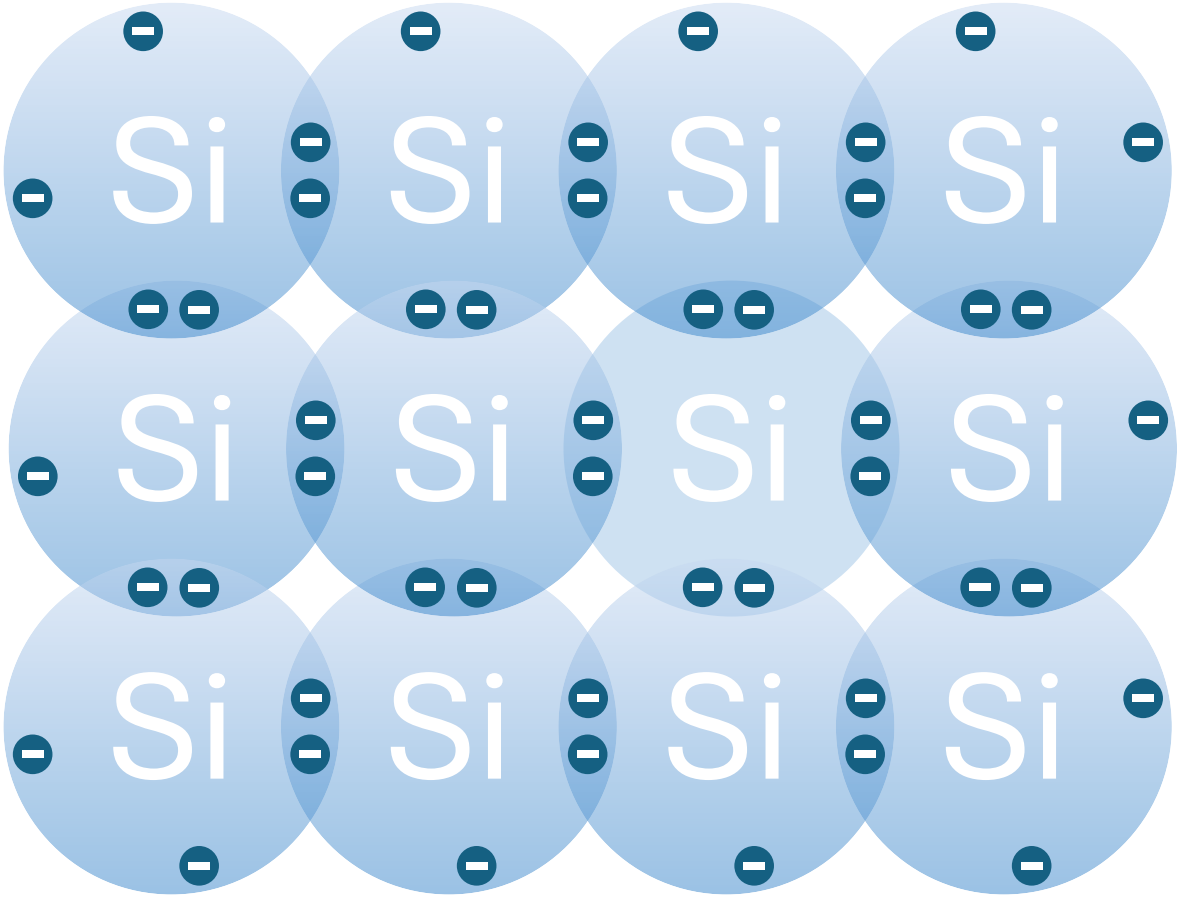
Silicon by Enricoros /
Silicone by OttawaAC / CC BY-SA 3.0
Silicon by Greg Robson / CC-BY-SA-2.0-UK
Periodic Table by Keith Enevoldsen / CC BY-SA 4.0
Argon by Pumbaa & Greg Robson / CC-BY-SA-2.0-UK
Silicone by Achim Hering / CC BY 3.0
Silicone by OttawaAC / CC BY-SA 3.0

Semiconductors



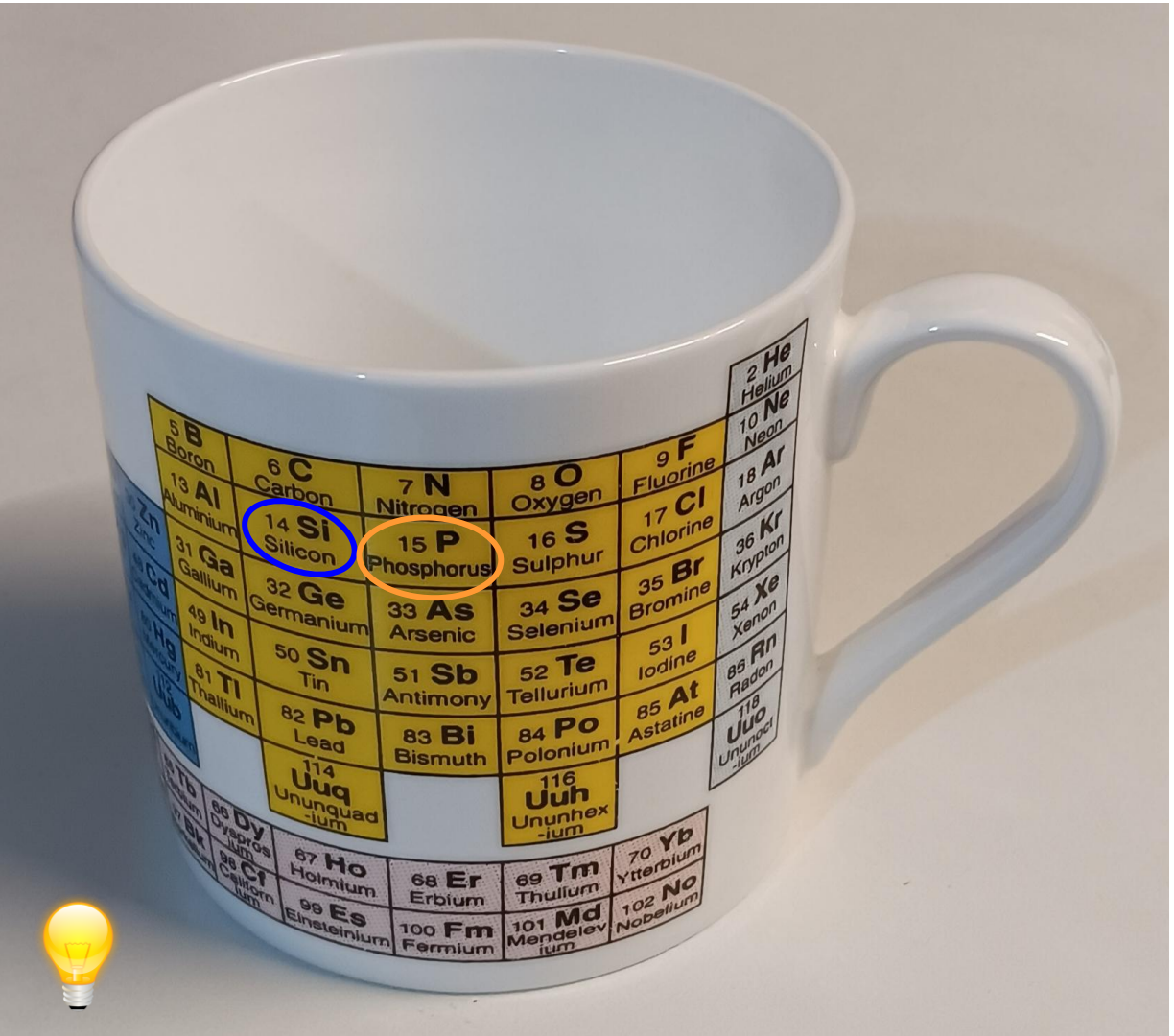


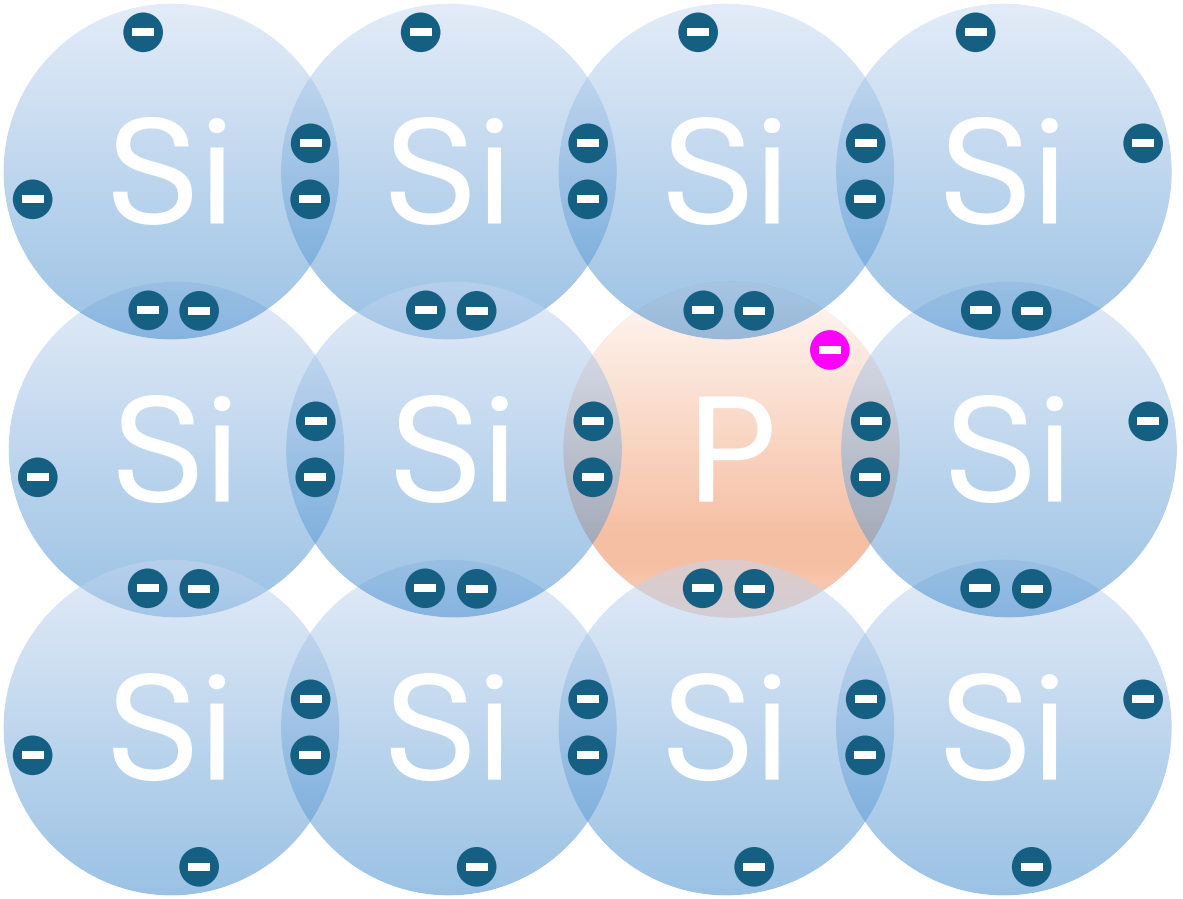
Mundane, stubborn, and profoundly disappointing

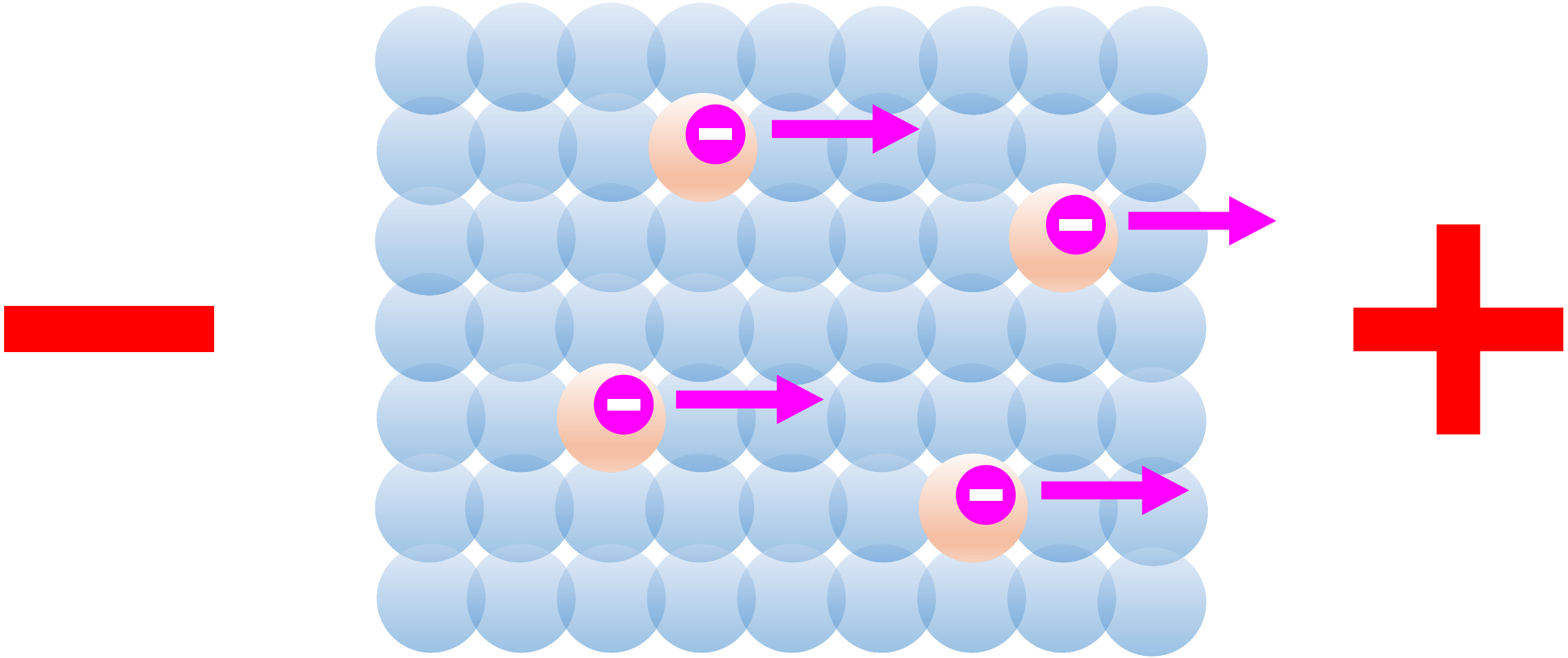


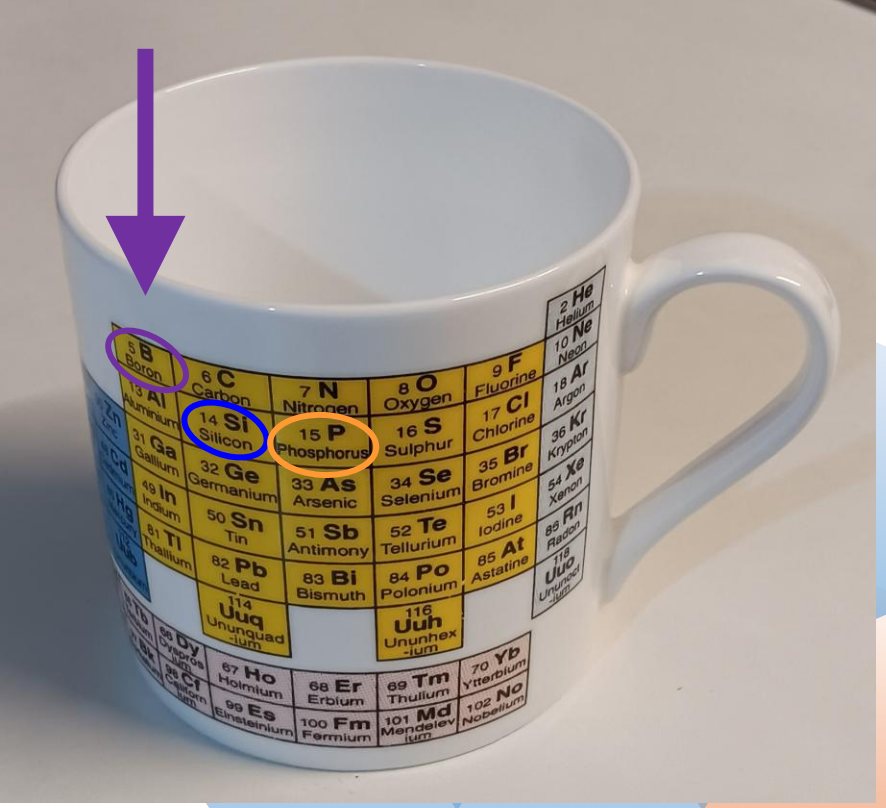
Semiconductors

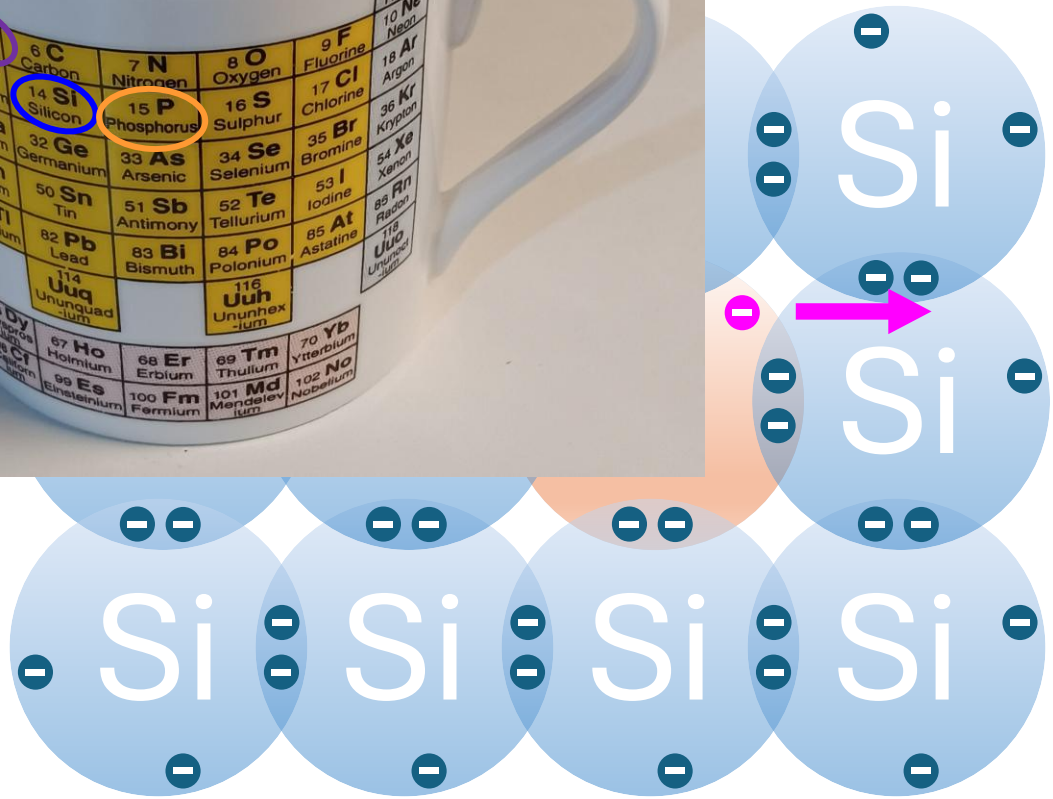
electron

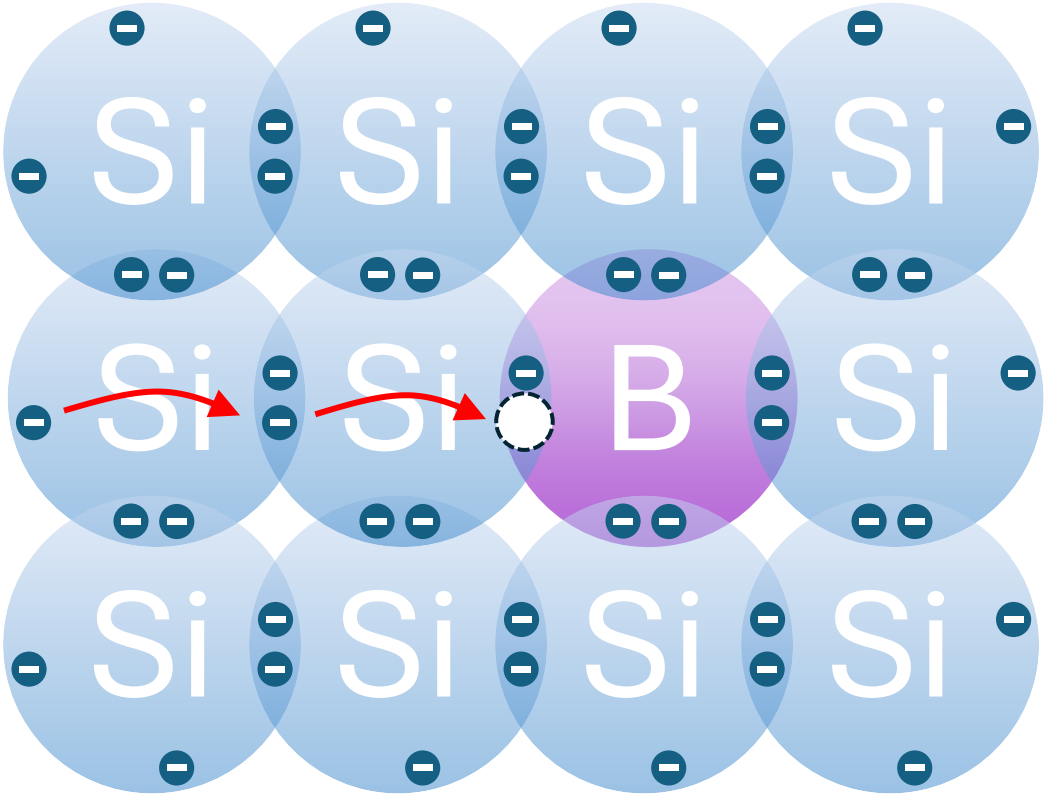








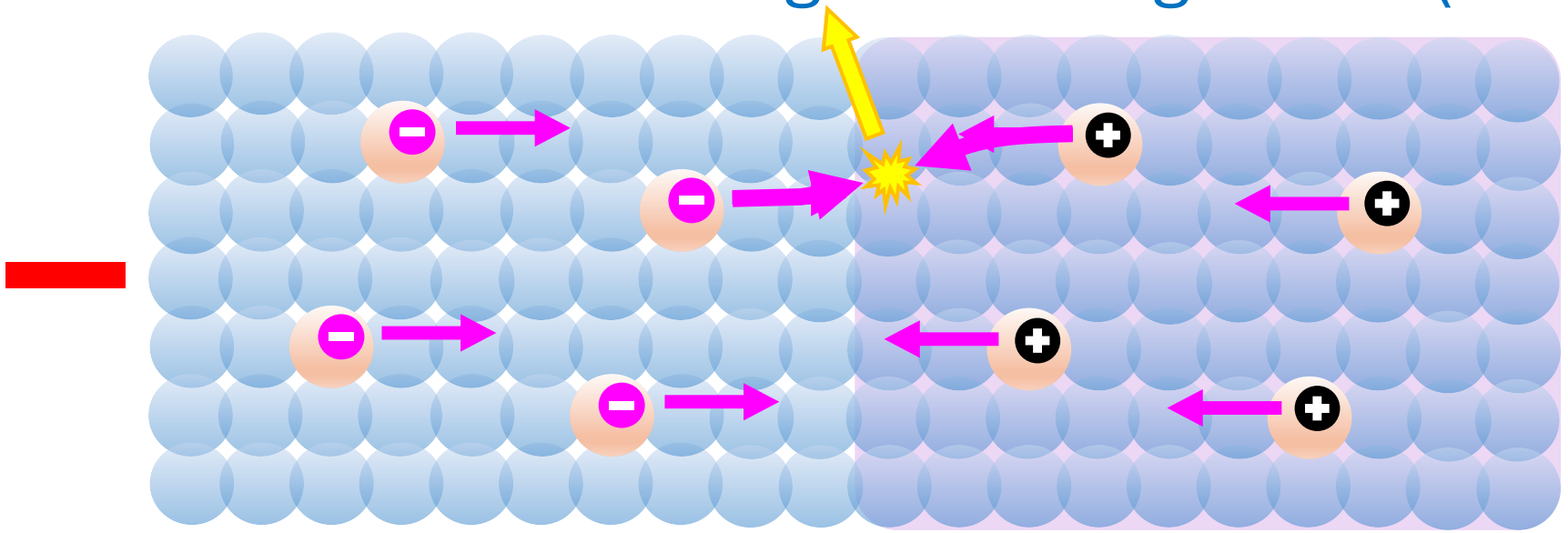




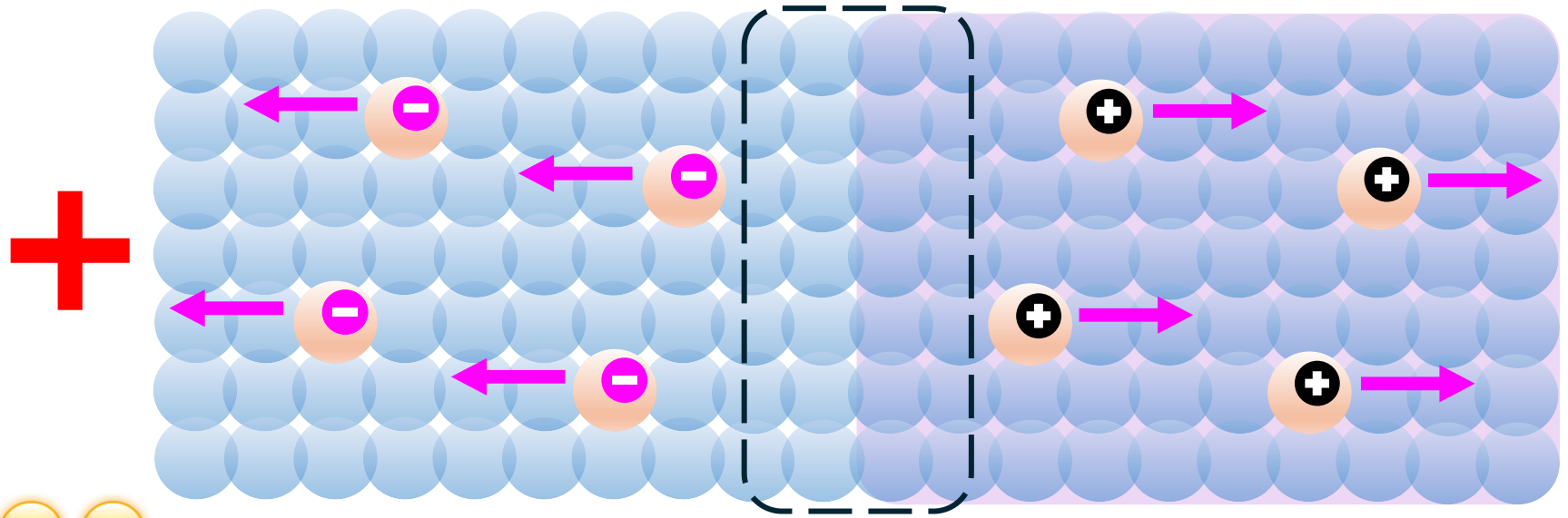
Electron →

← Electron hole

Light emitting diode (LED)

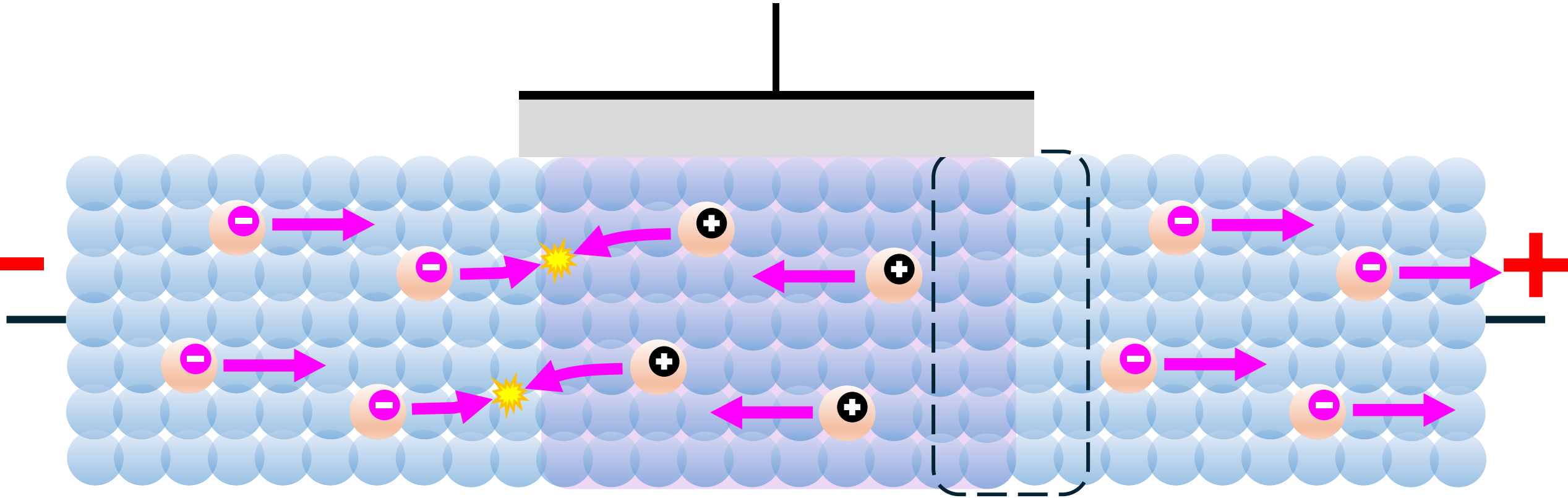


Diode

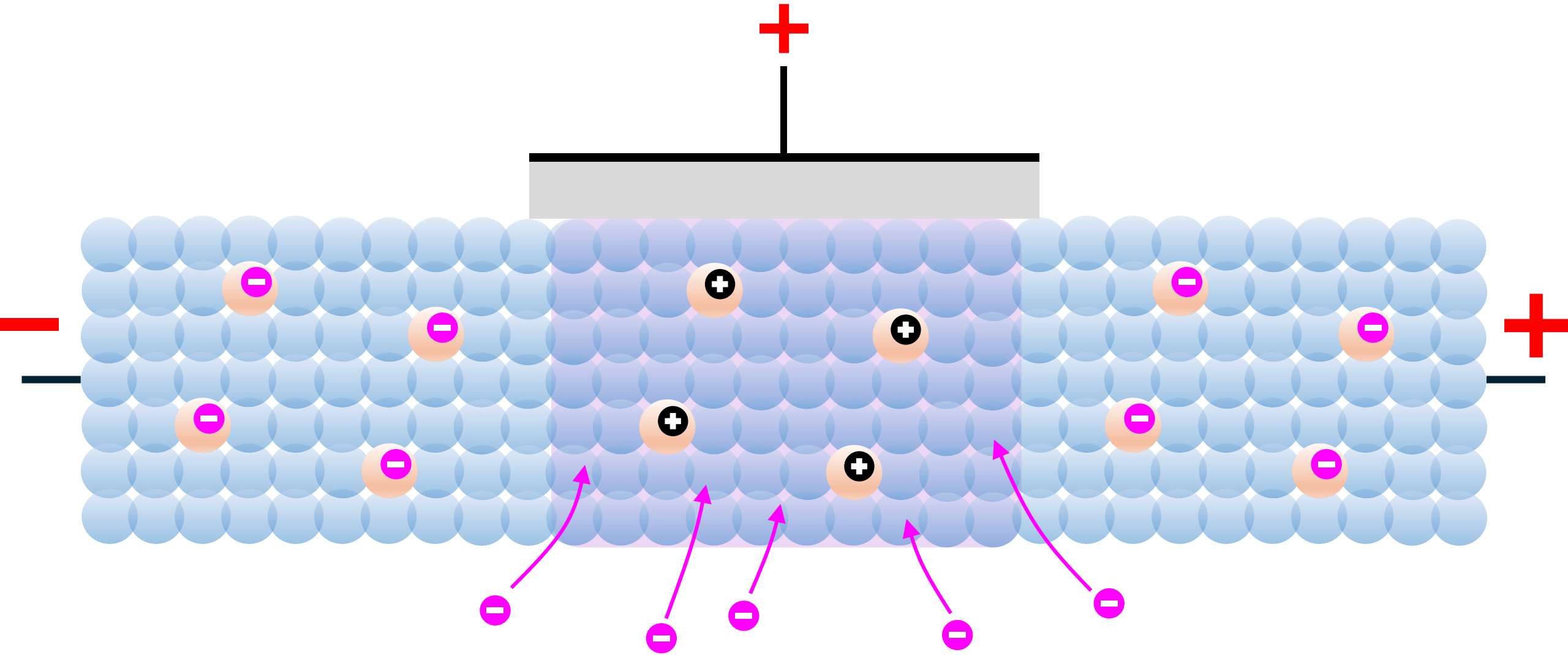


doesn't conduct

one way only!



Transistor

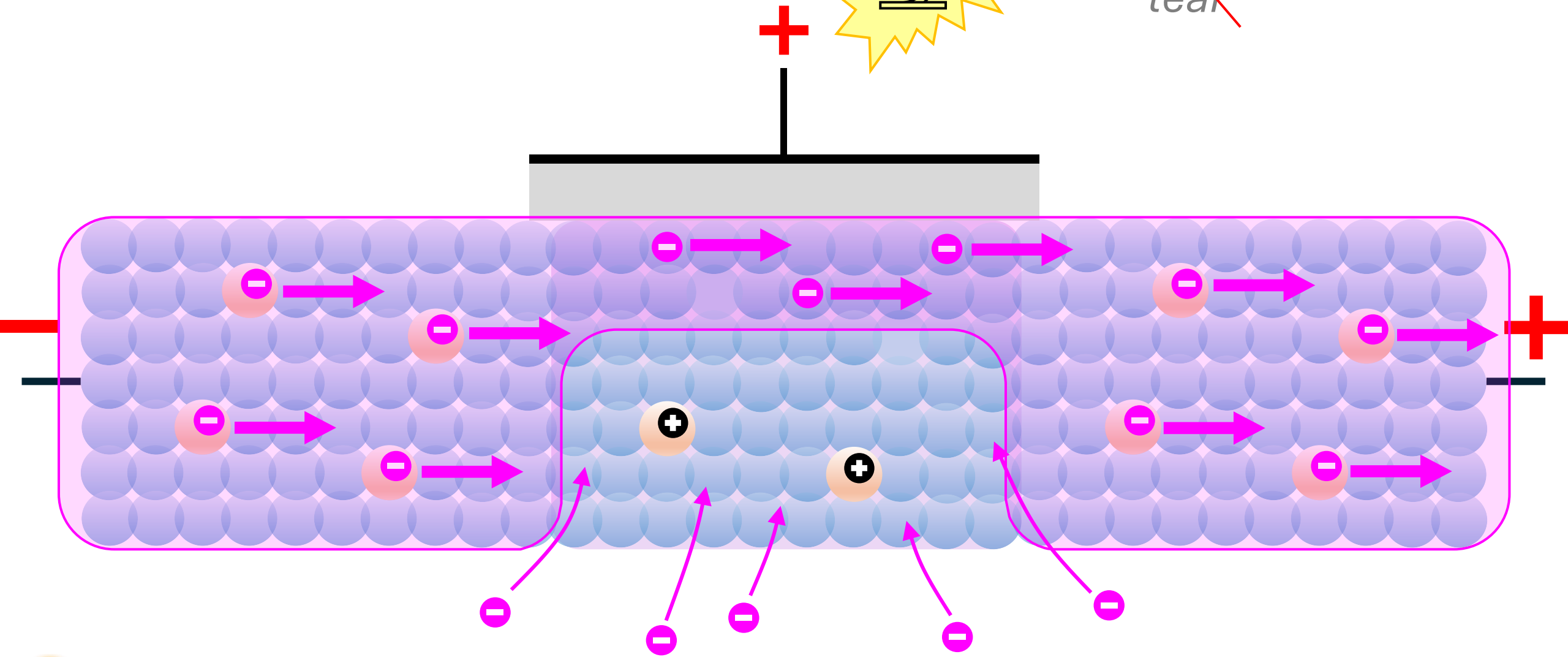


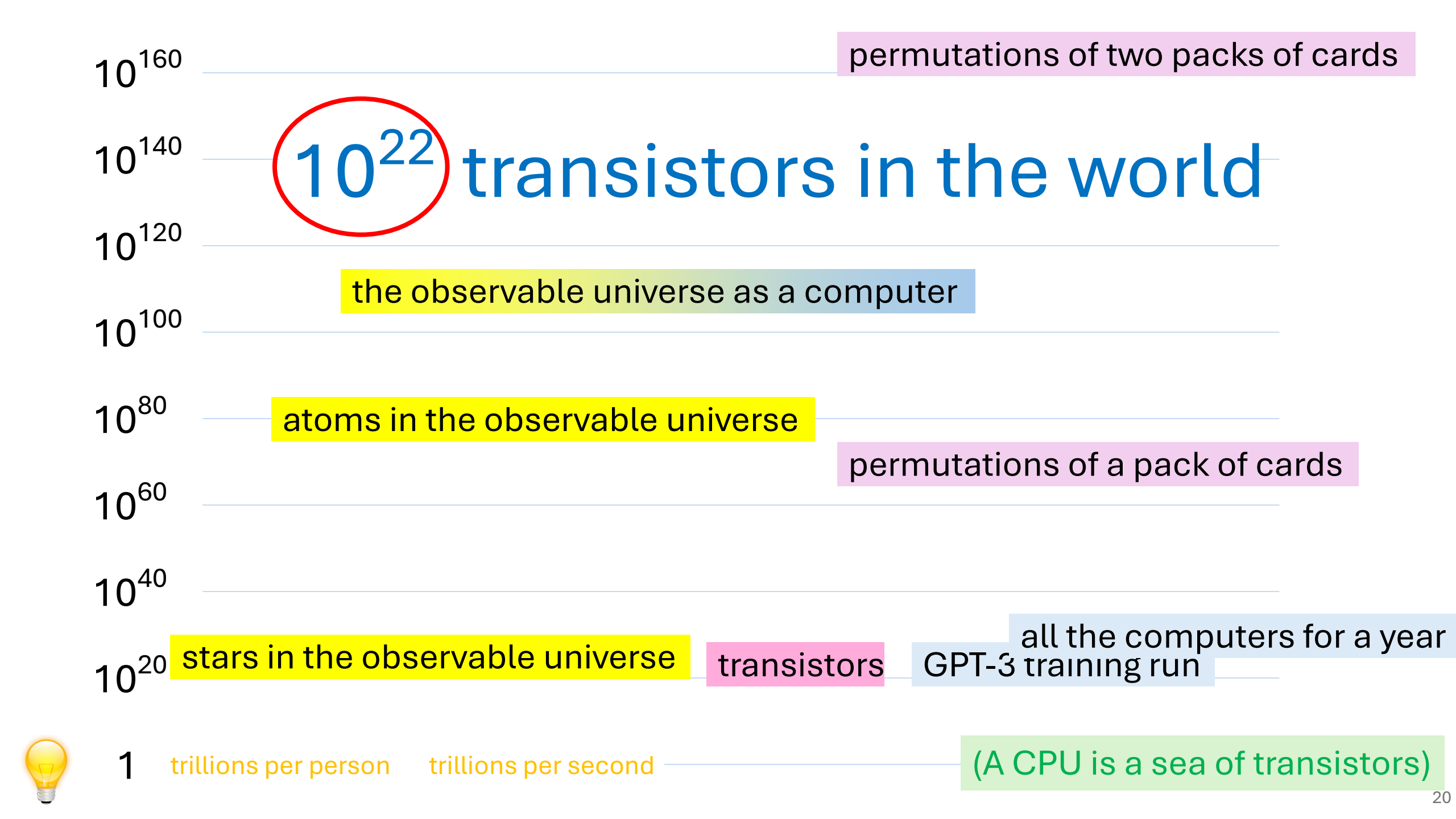
Transistor



wear and tear

on / off

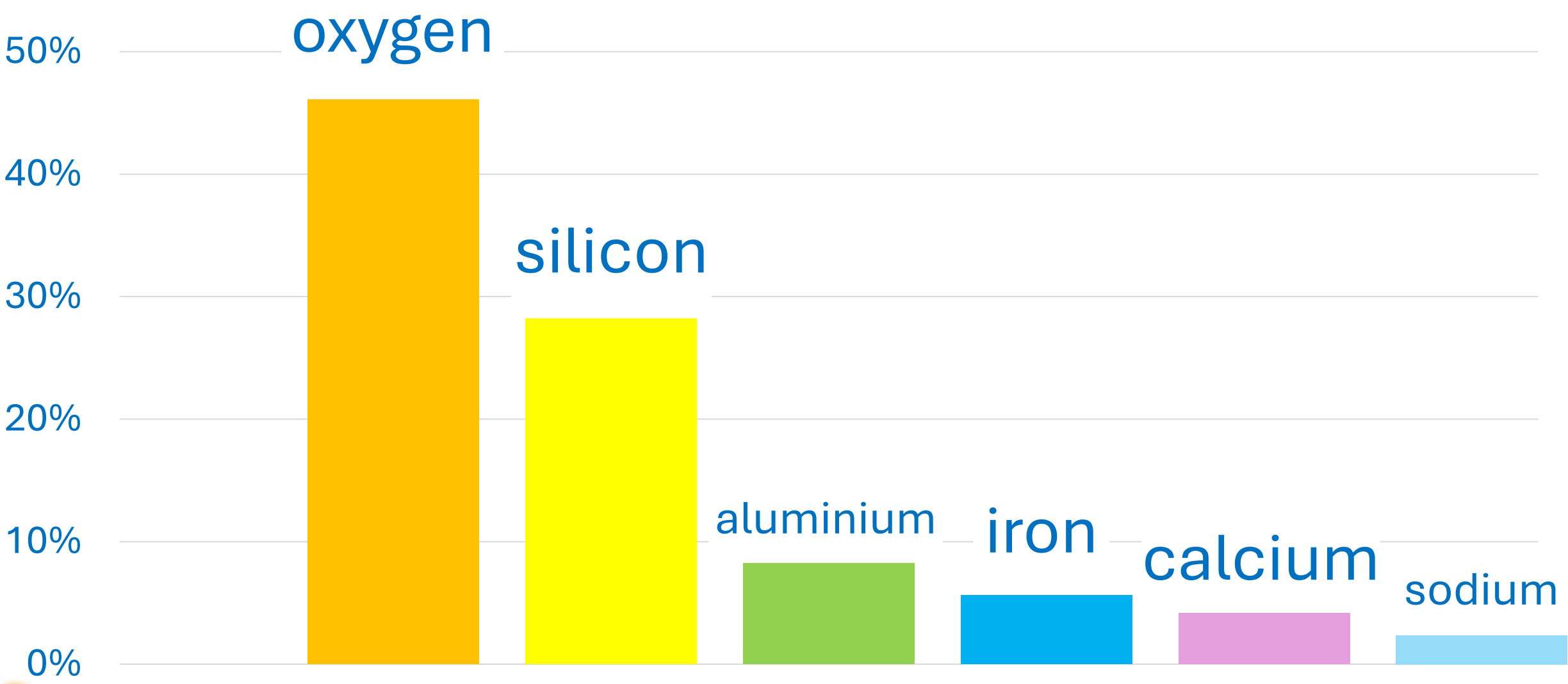



permutations of two packs of cards
10^{22} transistors in the world
10^{160}
10^{140}
10^{120}
10^{100}
10^{80}
10^{60}
10^{40}
10^{20}
1
the observable universe as a computer
atoms in the observable universe
permutations of a pack of cards
stars in the observable universe
transistors
GPT-3 training run
all the computers for a year
trillions per person
trillions per second
(A CPU is a sea of transistors)

silicon



Abundance of chemical elements in Earth's (continental) crust, by mass



https://en.wikipedia.org/wiki/Abundance_of_elements_in_Earth%27s_crust#List_of_abundance_by_element

Jan Czochralski



1915

sand



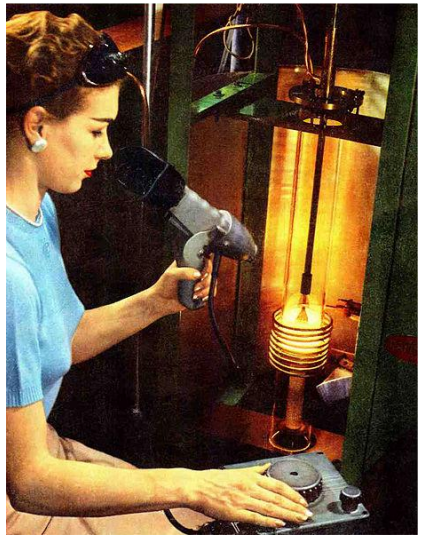
silicon



99.9999999%

quartz

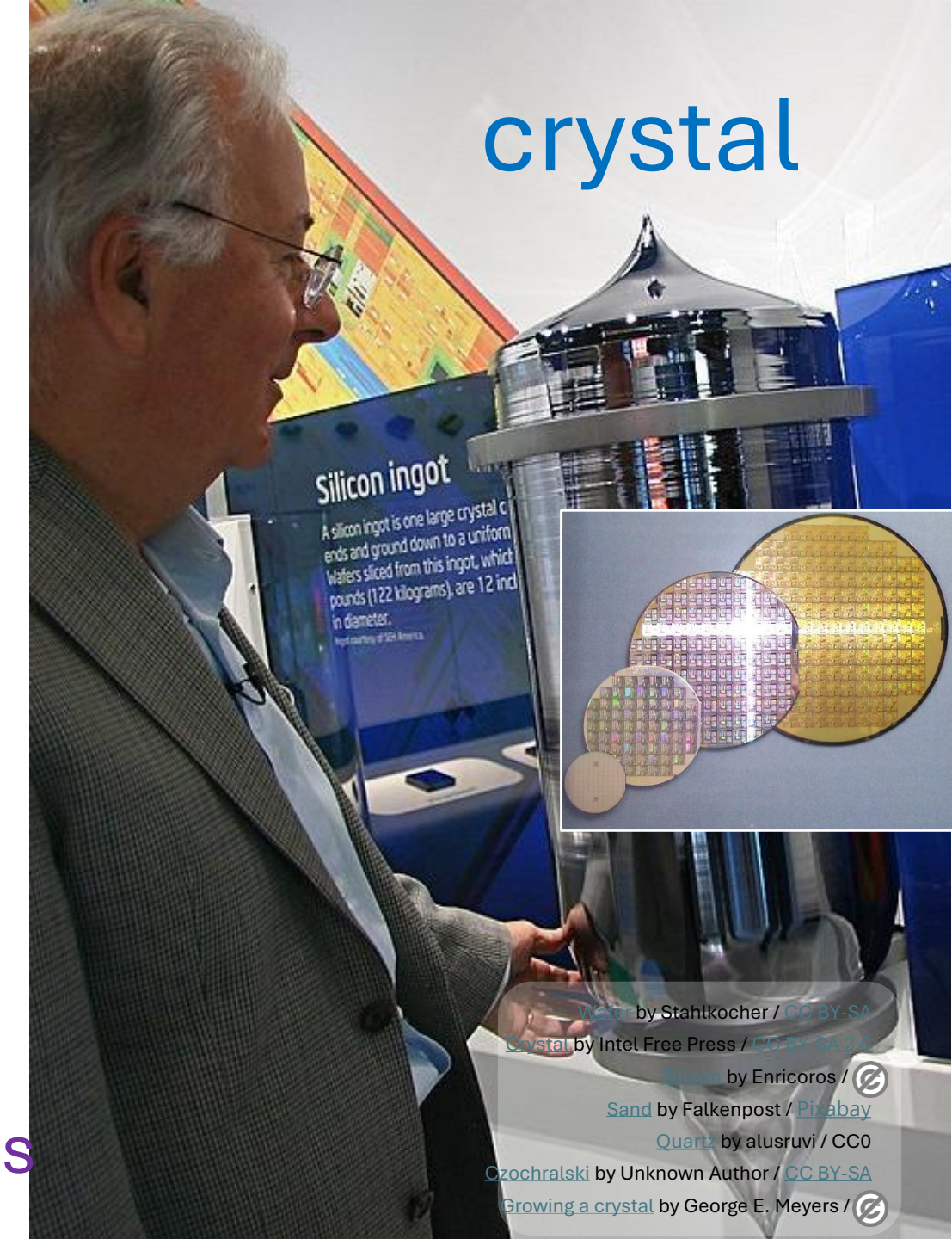




1956

Czochralski process

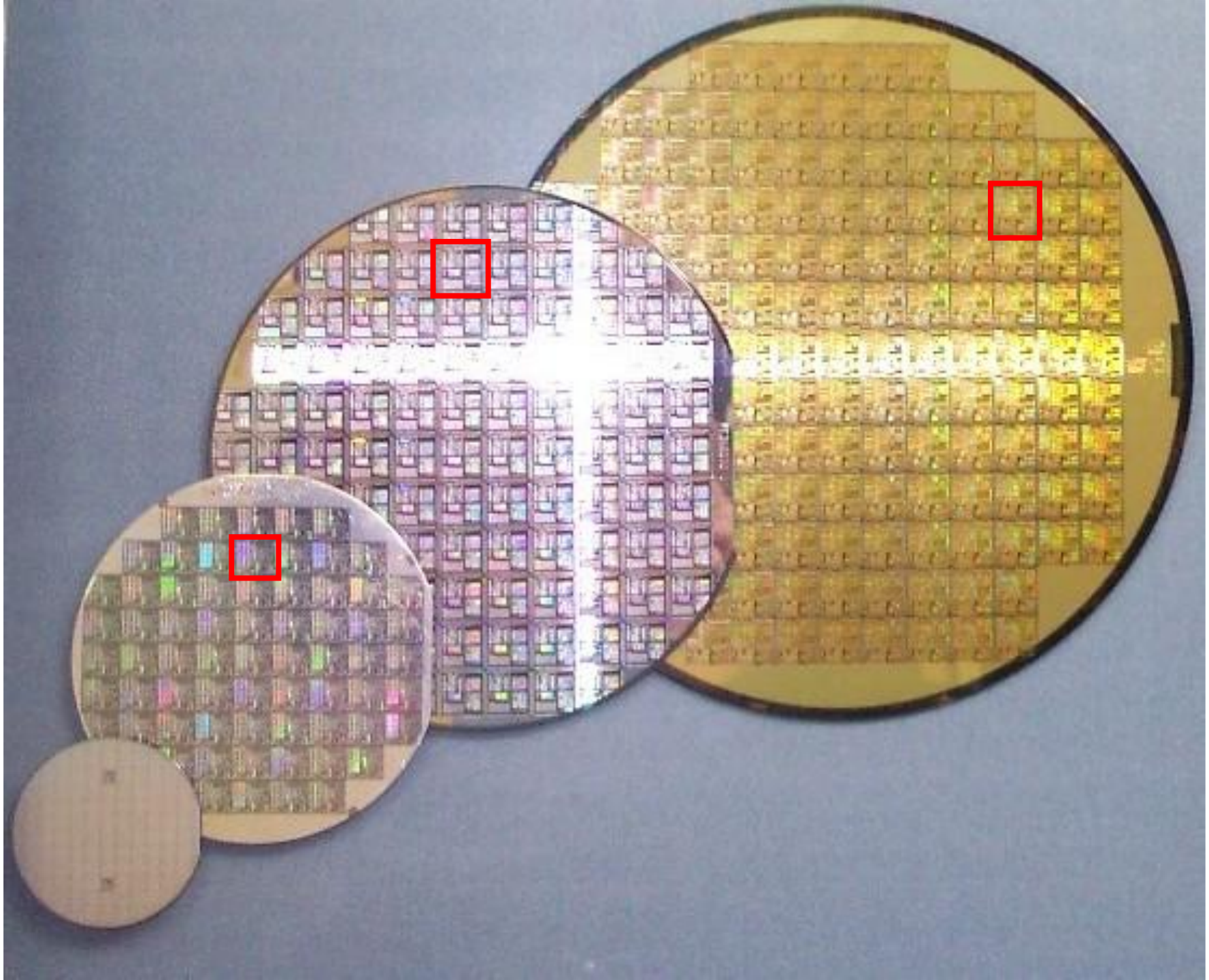
crystal



Wafer by Stahlkocher / CC BY-SA
Crystal by Intel Free Press / CC BY-SA 2.0
Silicon by Enricoros /
Sand by Falkenpost / Pixabay
Quartz by alusruvi / CC0
Czochralski by Unknown Author / CC BY-SA
Growing a crystal by George E. Meyers /

https://www.google.com/search?udm=2&q=wafer

(image search)



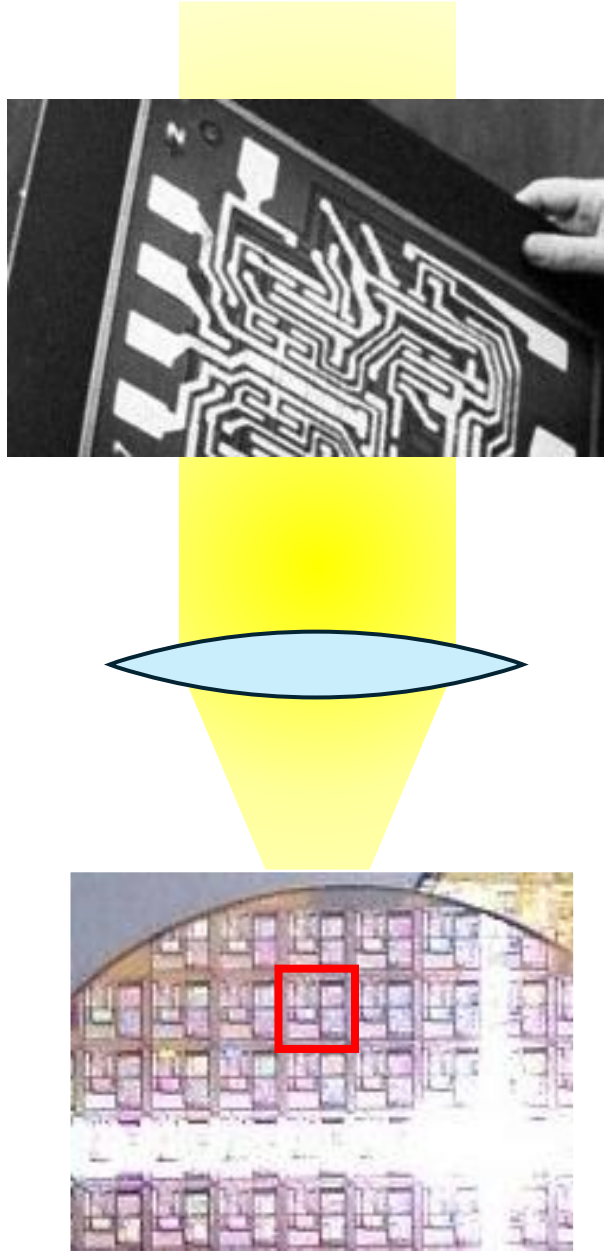


1959

Robert Noyce

Fairchild

Intel



Photolithography

Etching

Chemical vapour deposition

Robert Noyce by Intel Free Press / CC BY-SA 2.0

https://en.wikipedia.org/wiki/Semiconductor_fabrication_plant

Apple, Nvidia, ARM, ... *fabless*

TSMC – Taiwan *makes the chips*

semiconductor

ASML – Netherlands *makes the machines that make the chips*

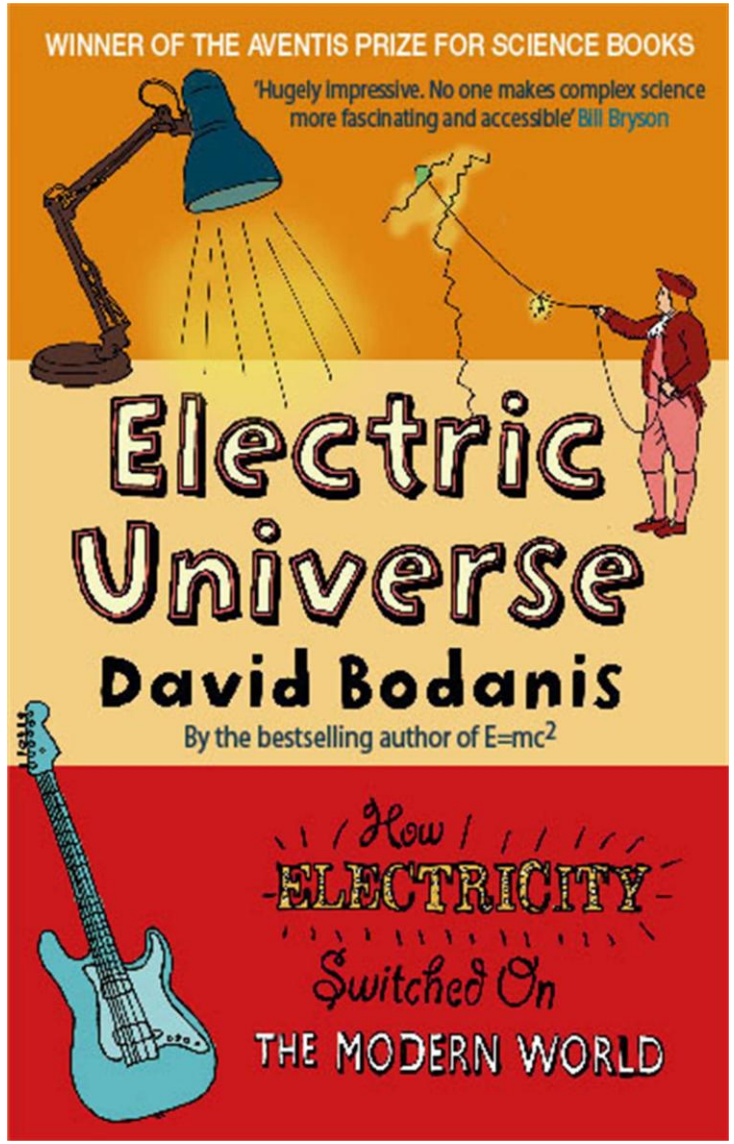
https://www.google.com/search?udm=2&q=chip+fab

https://www.google.com/search?udm=2&q=ASML





Lilienfeld
1926



Bardeen Shockley Brattain



1947

Lilienfeld / CC0
Bipolar Junction Transistor / CC0

Long-term memory

Magnetic tape



Hard disk drive



Short-term memory

RAM

Random-access memory

Memory



Short-term memory

Central-processing unit

CPU

+ − × ÷

RAM

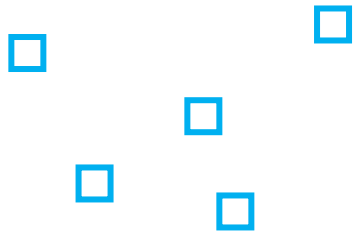
Random-access memory

Memory

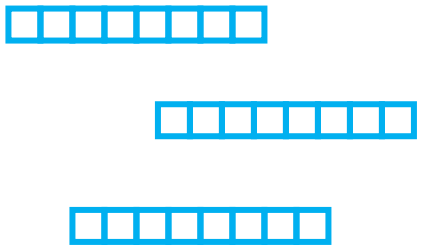


Memory

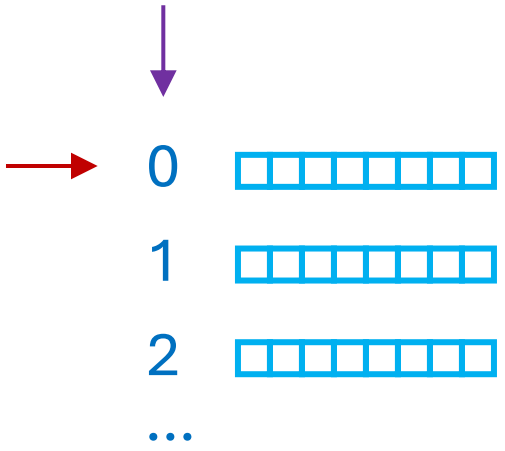
bits



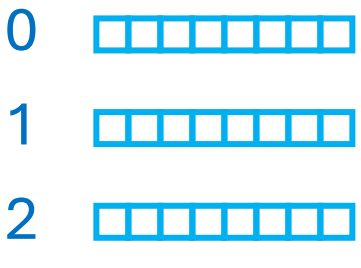
bytes



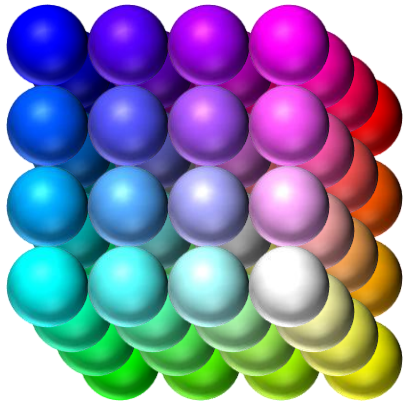
addresses



big



flat



森 β ê

א ぶ

Short-term memory

Central-processing unit

CPU

+ − × ÷

RAM

Random-access memory

Memory





simple

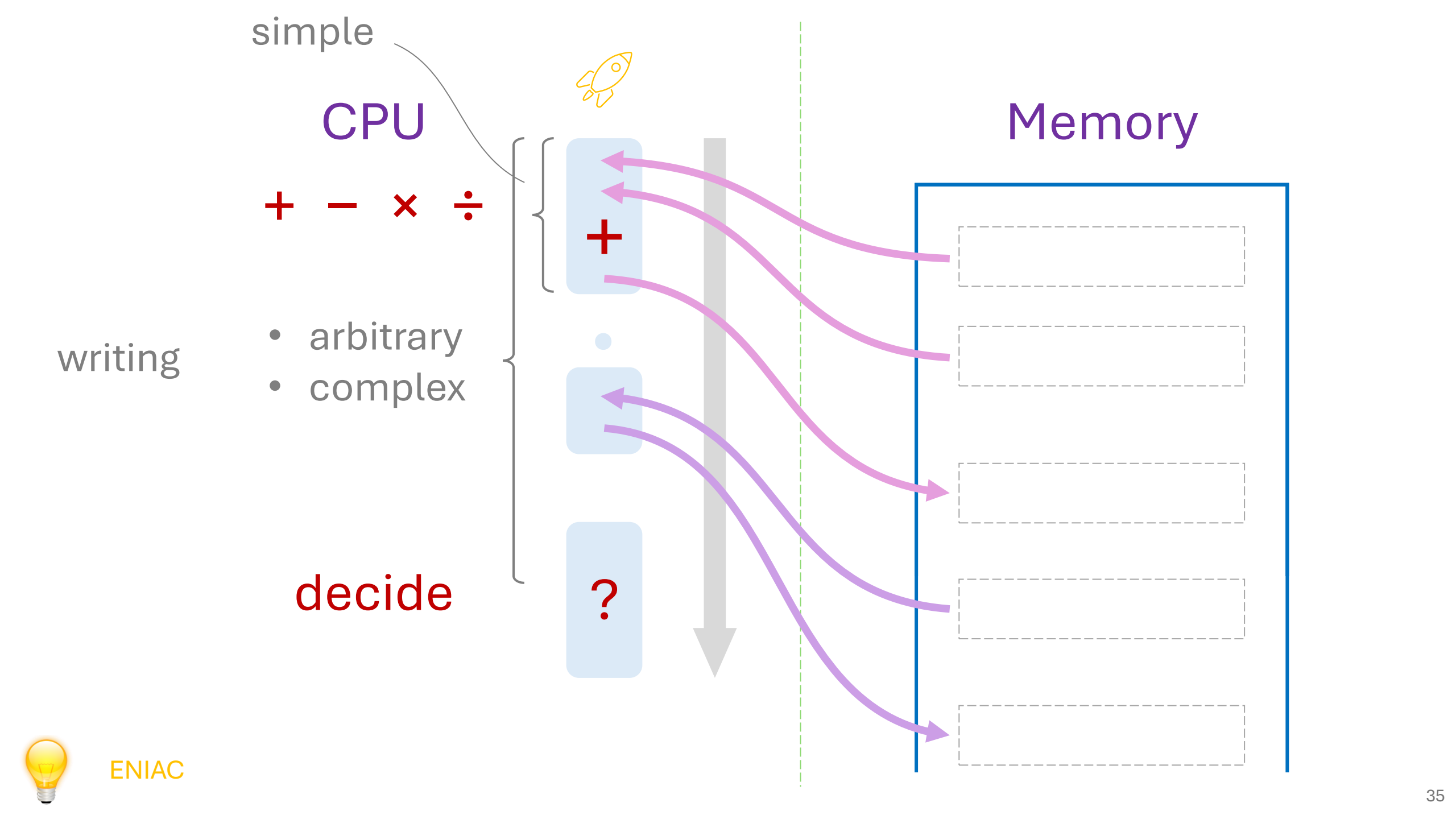
CPU

+ − × ÷

writing

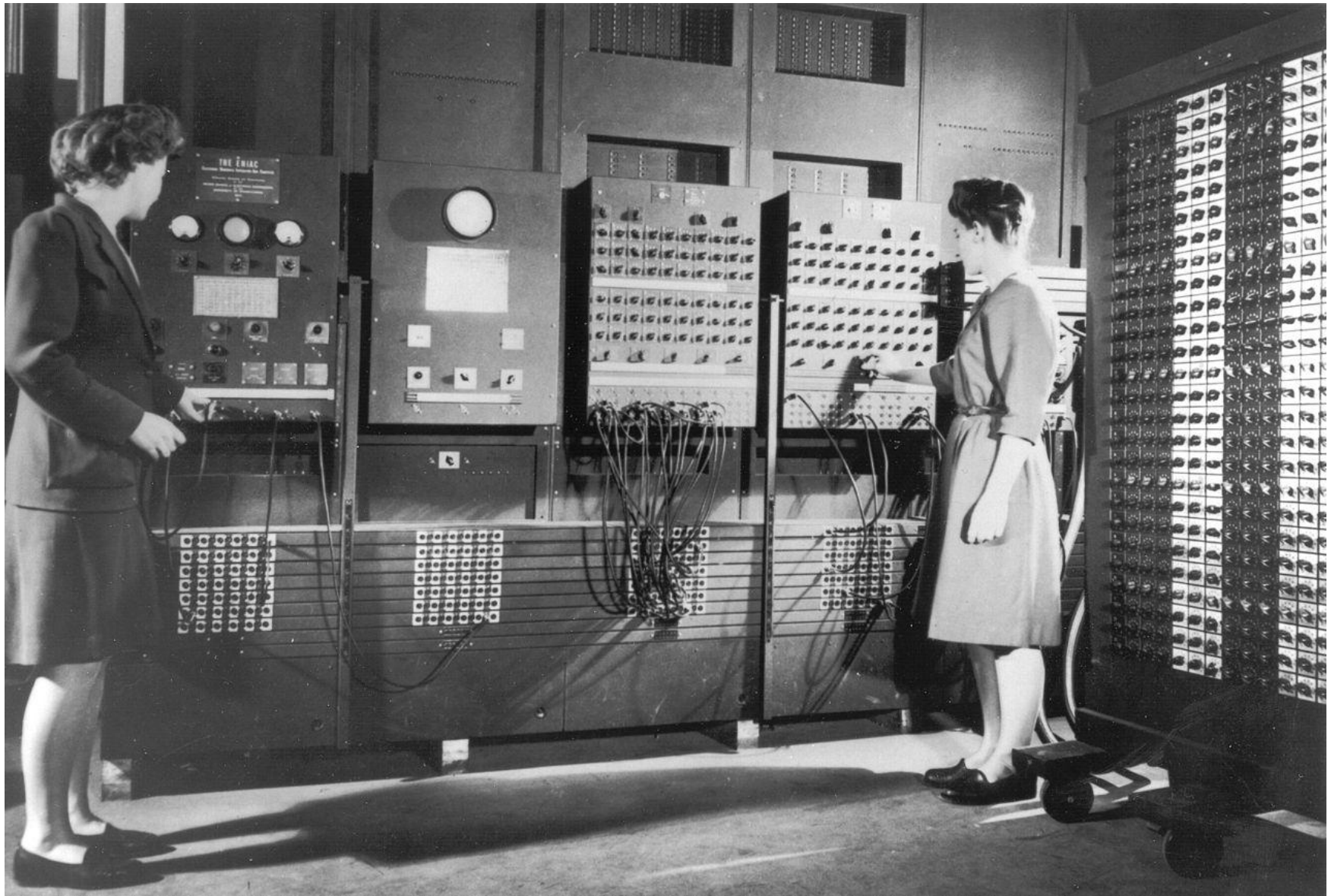
- arbitrary
- complex

Memory



ENIAC

Programming the ENIAC



1945

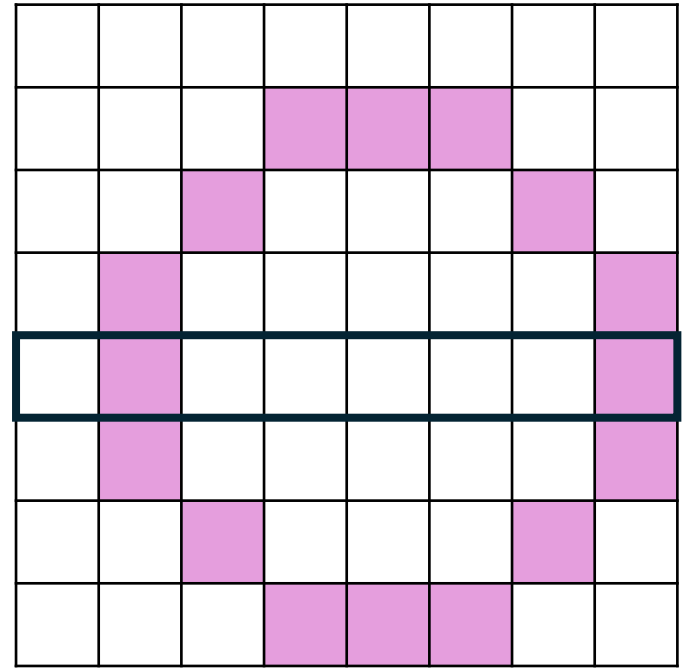
01000001 ‘A’

01000001 65

64 32 16 8 4 2 1

01000001





01000001 *a computer instruction*

or part of one

Computer

Fast

Deterministic

Understandable

Unbelievably good at following instructions

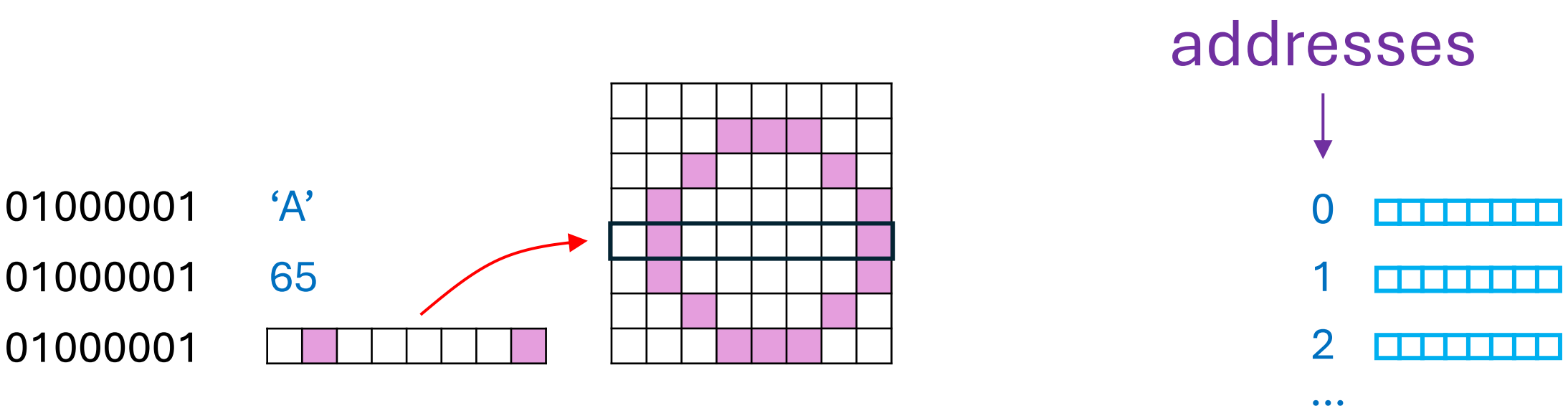
Brain

Parallel

Low energy

Local memory

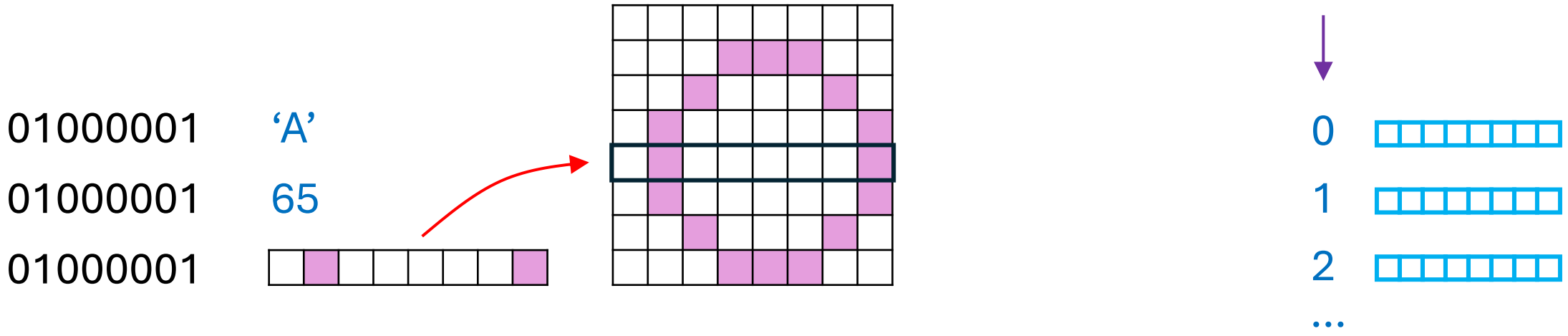




L	o	n	d	o	n			8,866,180
Y	o	r	k					141,685

?

addresses



L	o	n	d	o	n			8,866,180
Y	o	r	k					141,685
C	a	m	b	r	i	d	g	1,694,645,811



big enough?



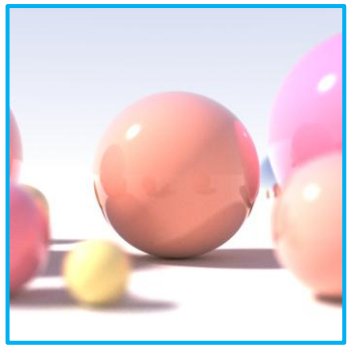
Name by Rob Koster / CC BY-SA 3.0

Parallel computers





Ray tracing



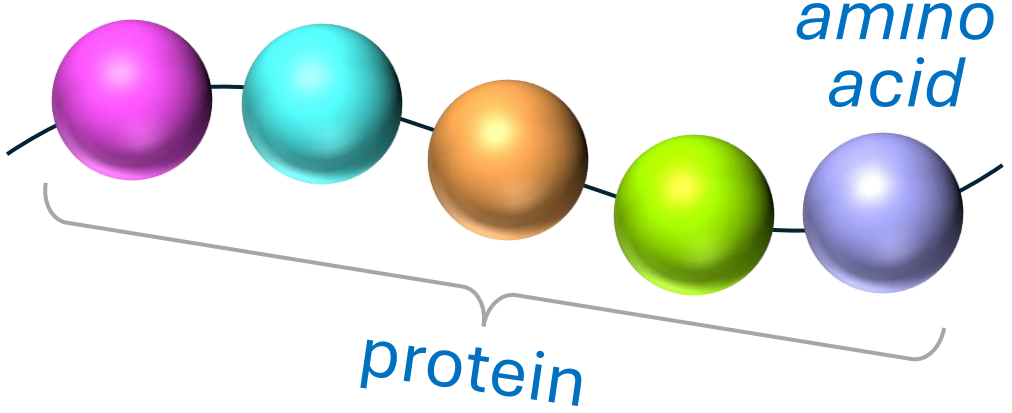
a + b

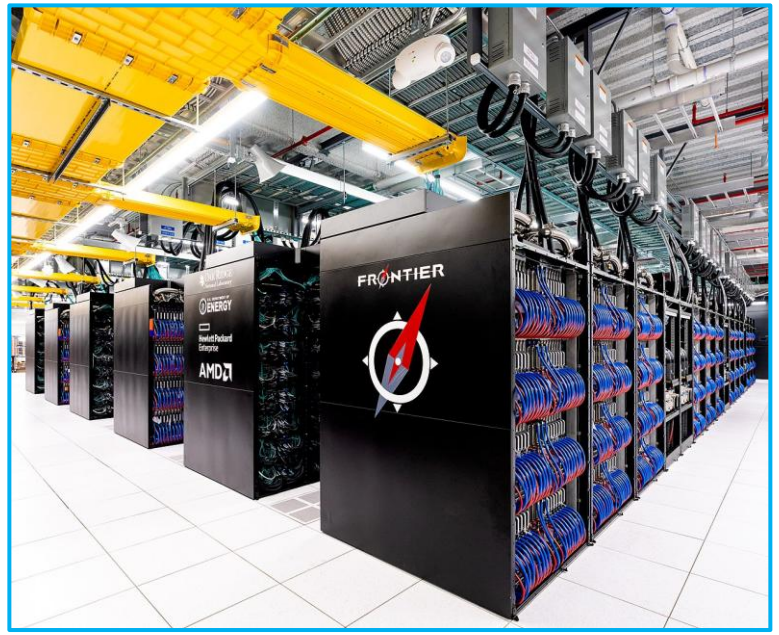
c + d

parallel

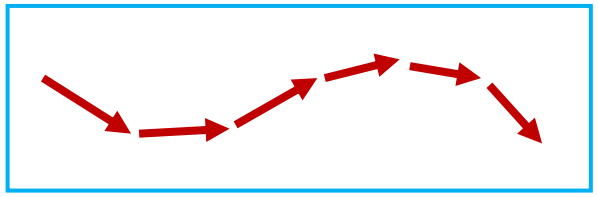
Searching







x + y + z ✗





Frontier by OLCF / CC BY 2.0
Ray tracing by Tim Babb / CC BY-SA 4.0

Sequential

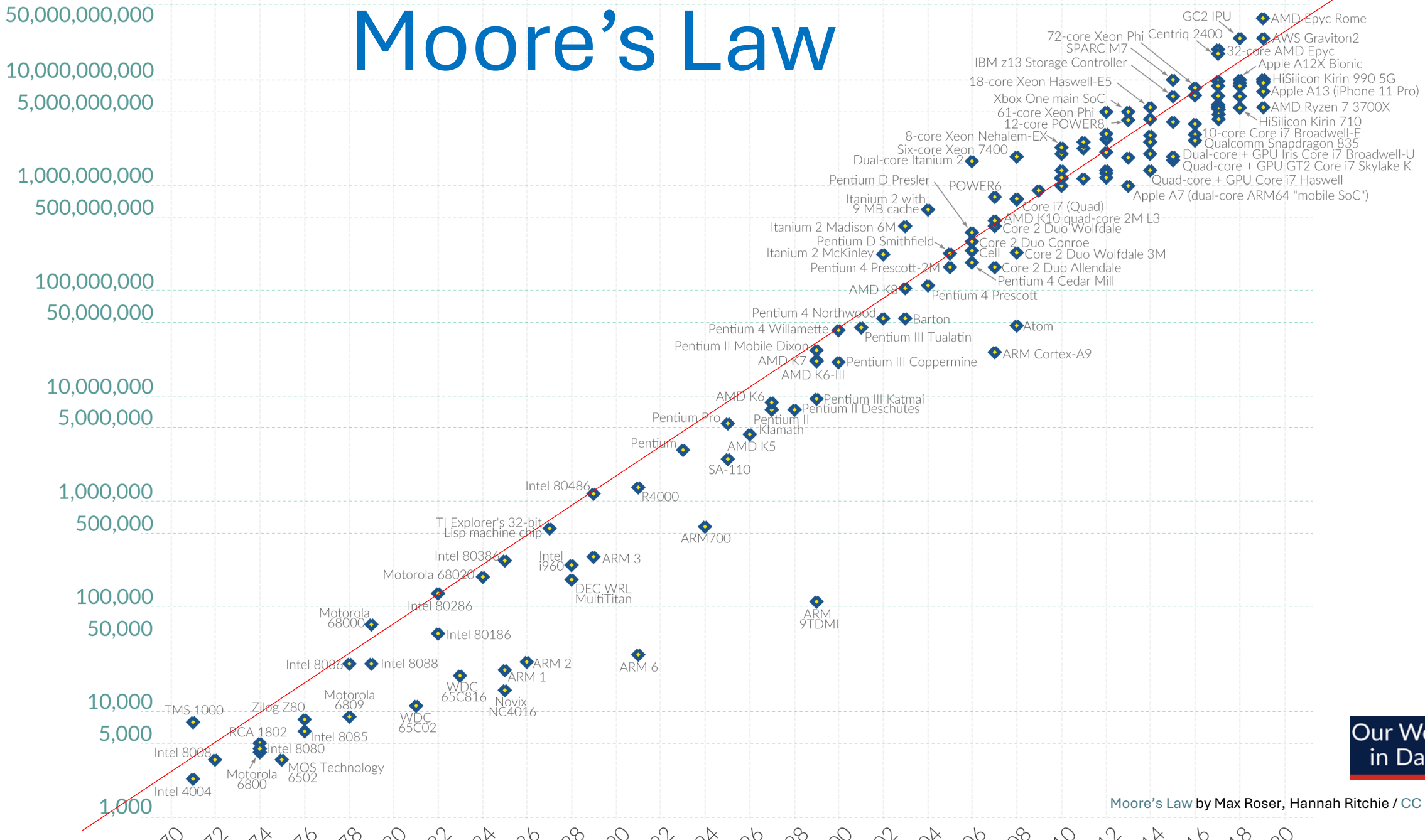
- Our conscious thoughts
- 20th century computers

Parallel

- Our subconscious
- The world



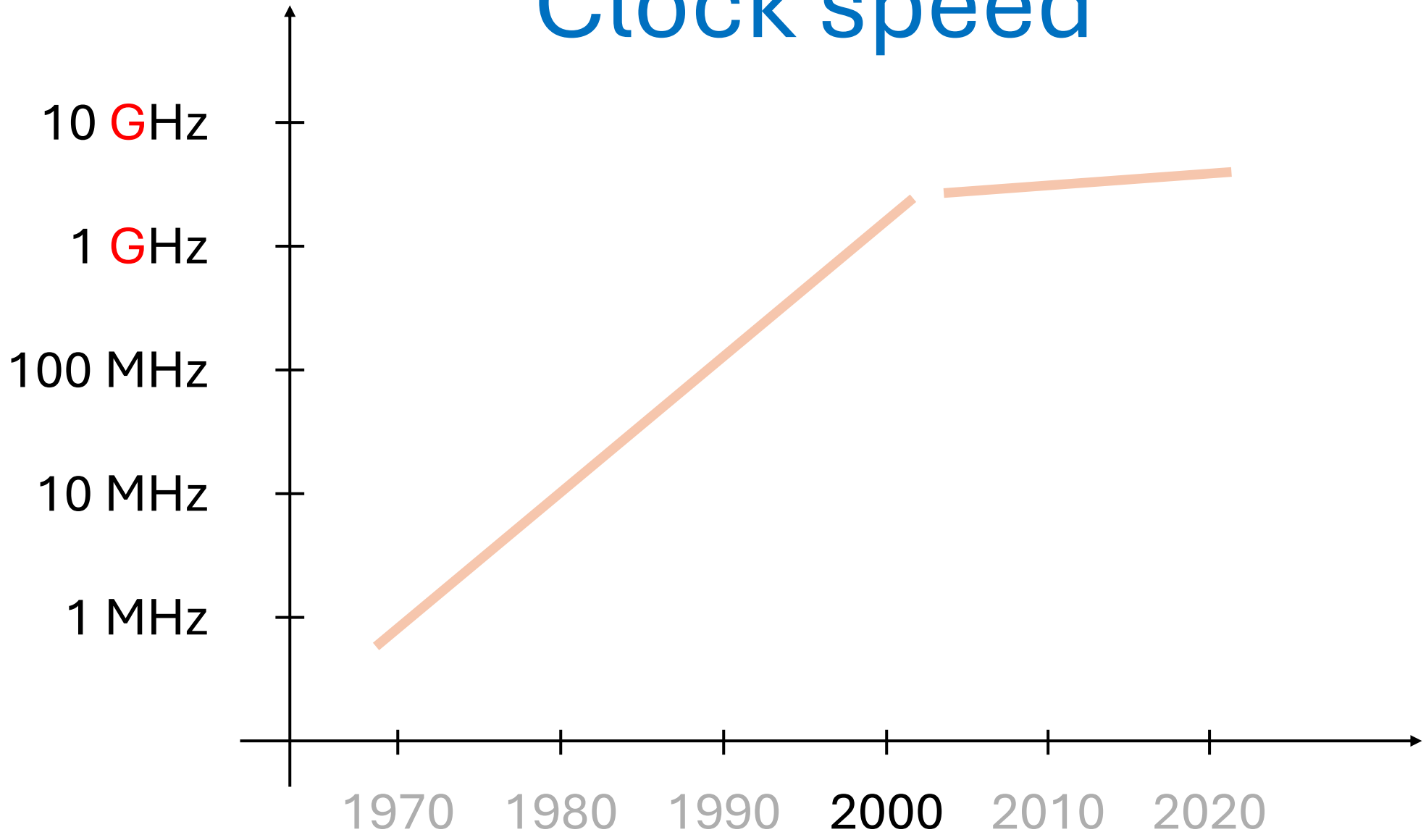
Moore's Law

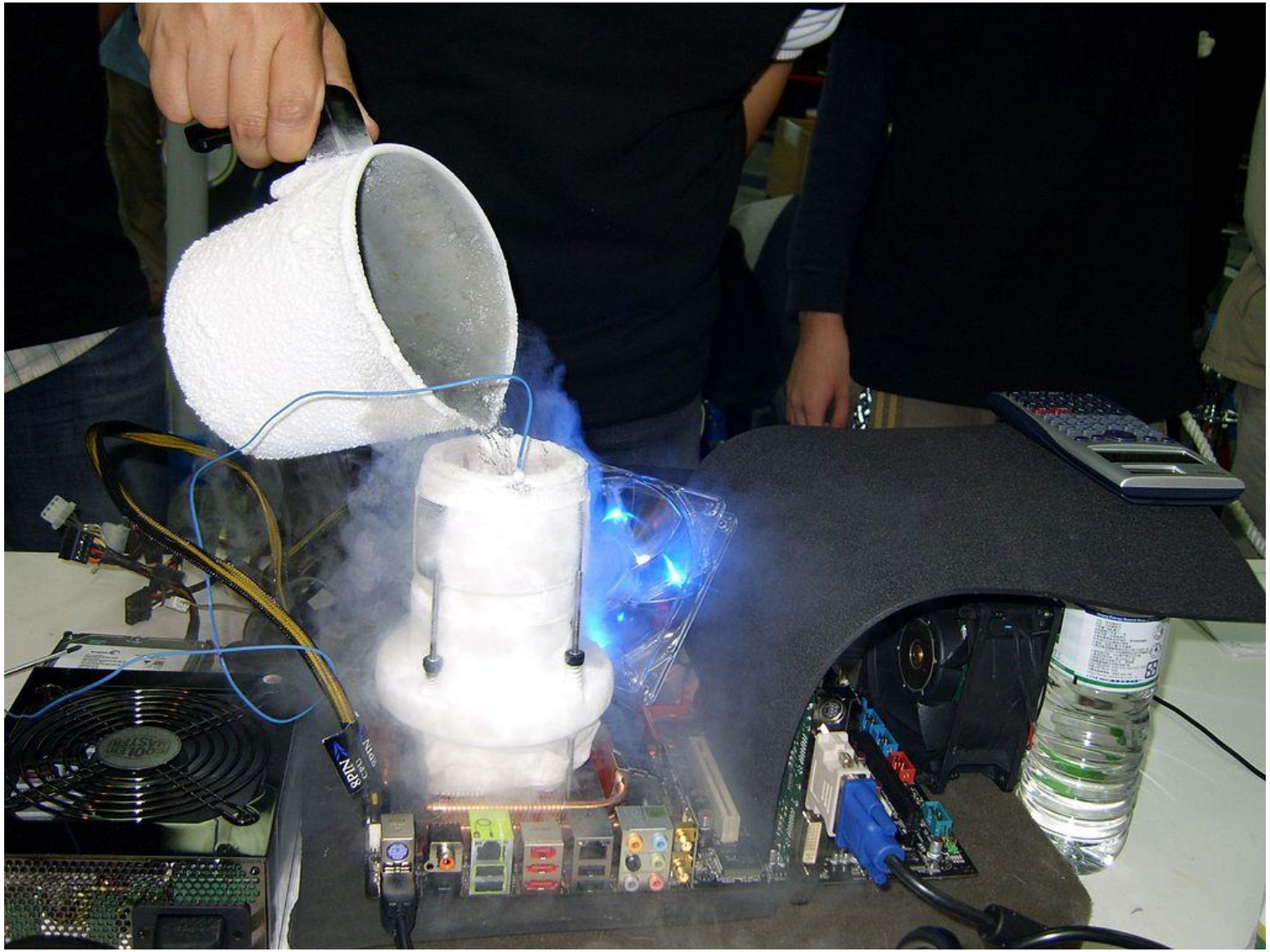


Our World in Data

Moore's Law by Max Roser, Hannah Ritchie / CC BY 4.0

Clock speed

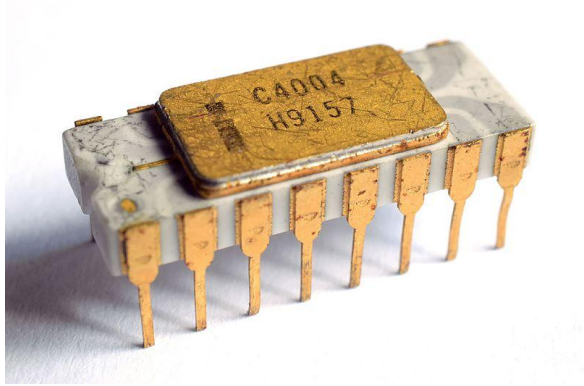


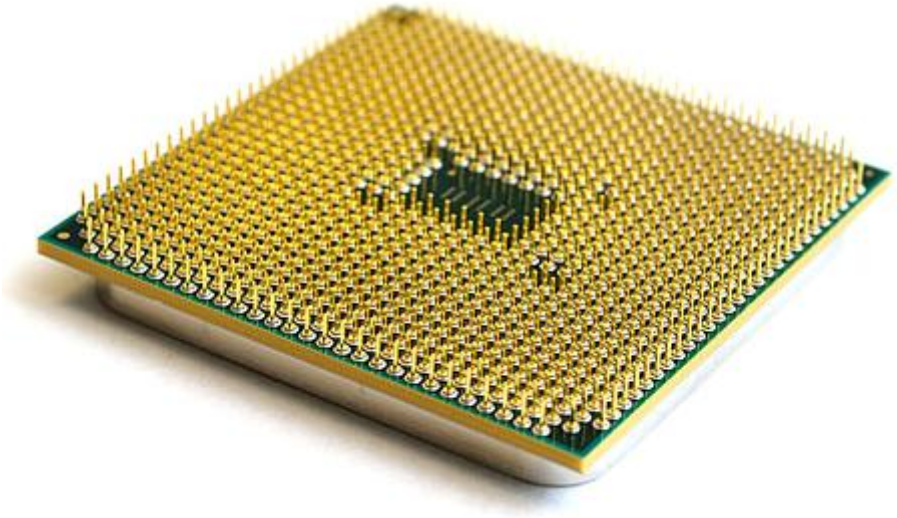


−196 °C

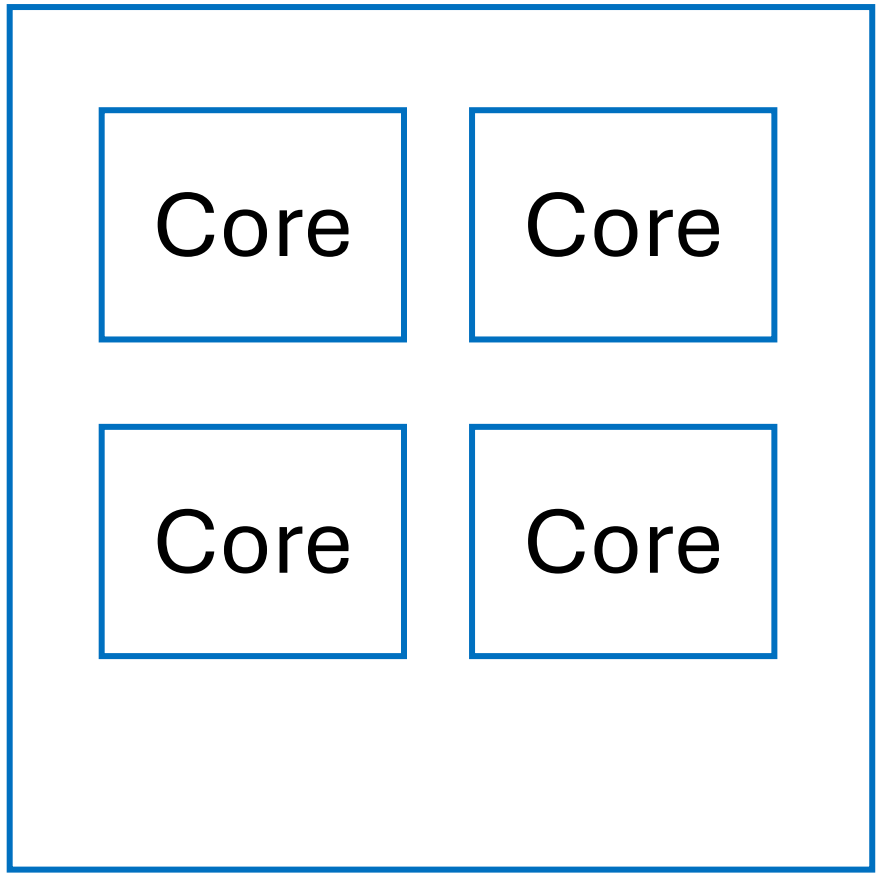


Overclocking by Rico Shen / CC BY-SA 3.0





Quad-core Chip

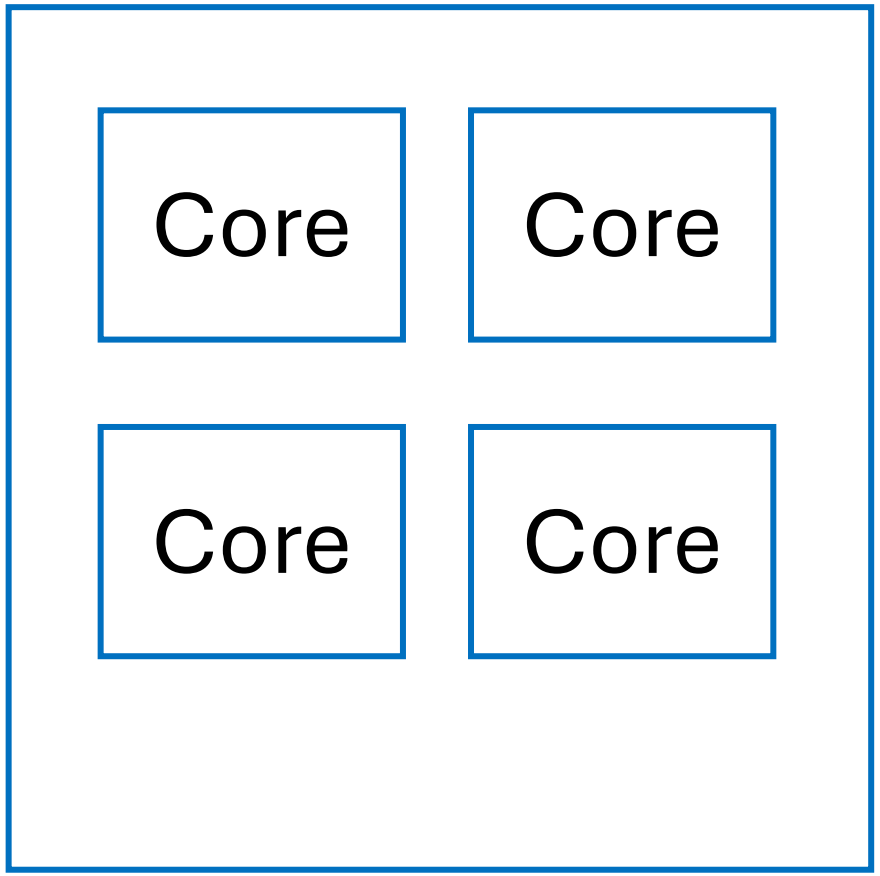




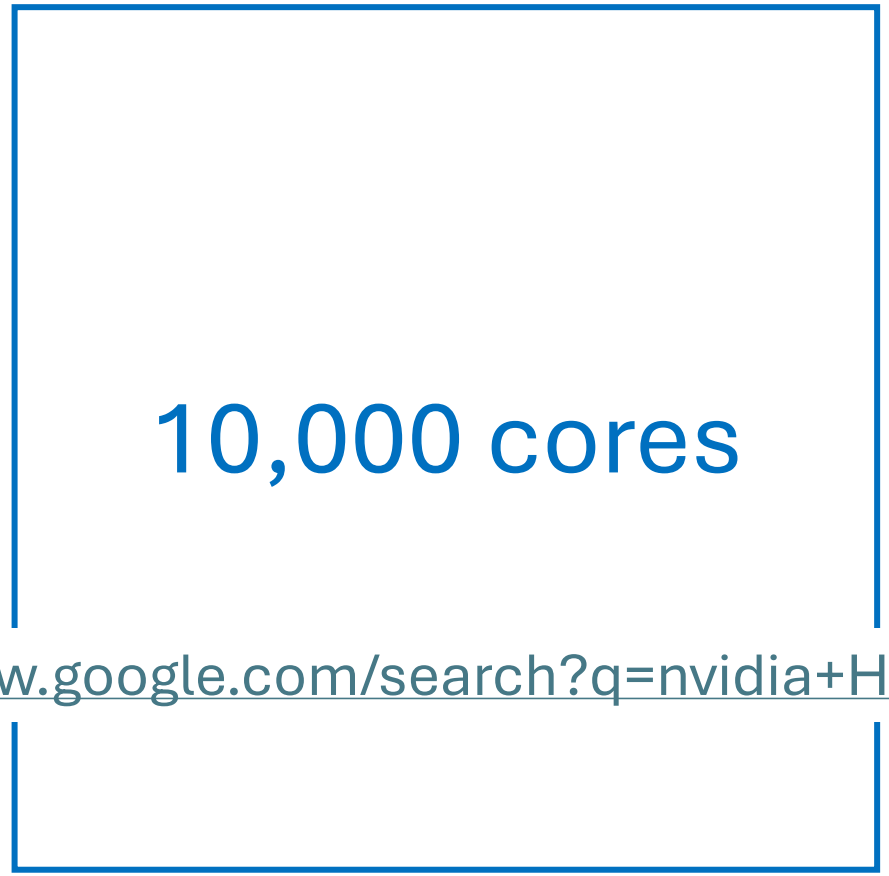
CPU from PickPik

4004 by Stelo.xyz, Pttn, or Thomas Nguyen / CC BY-SA 4.0

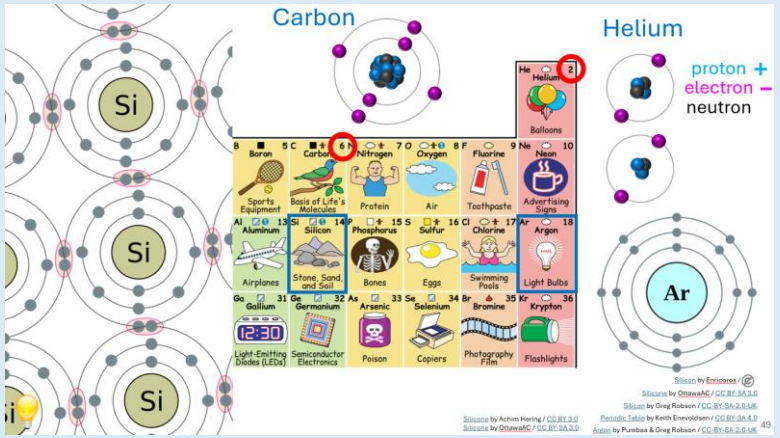
Quad-core CPU

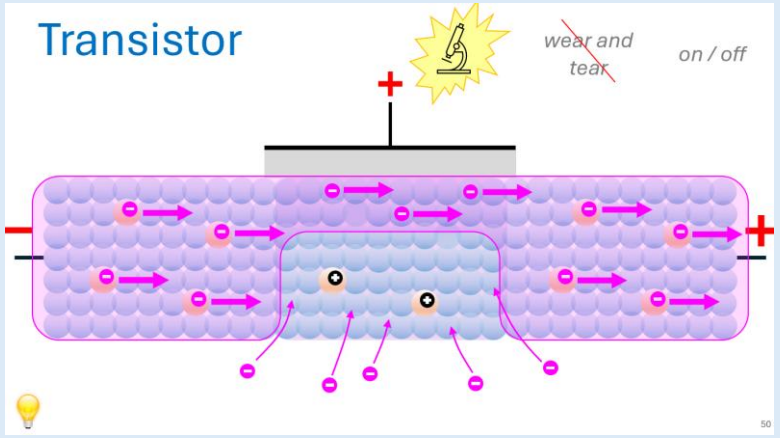


GPU

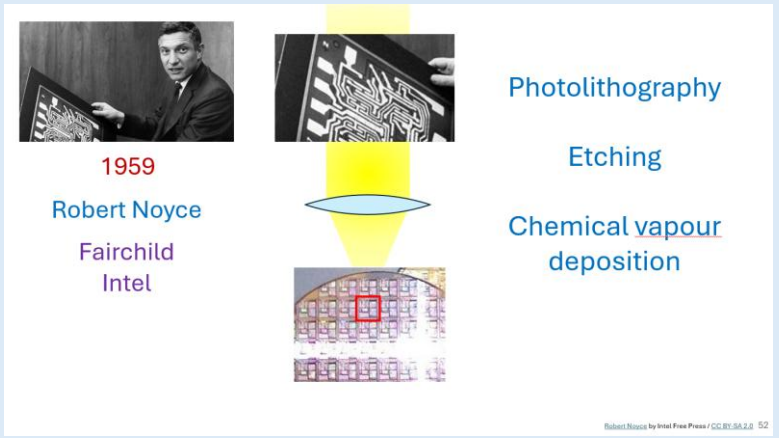


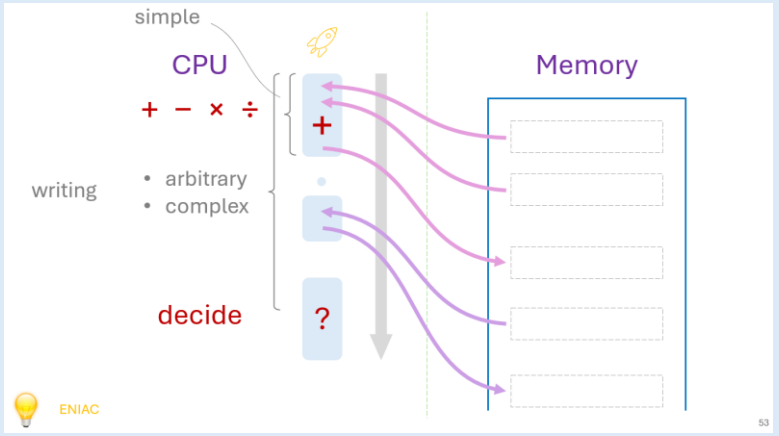
https://www.google.com/search?q=nvidia+H100

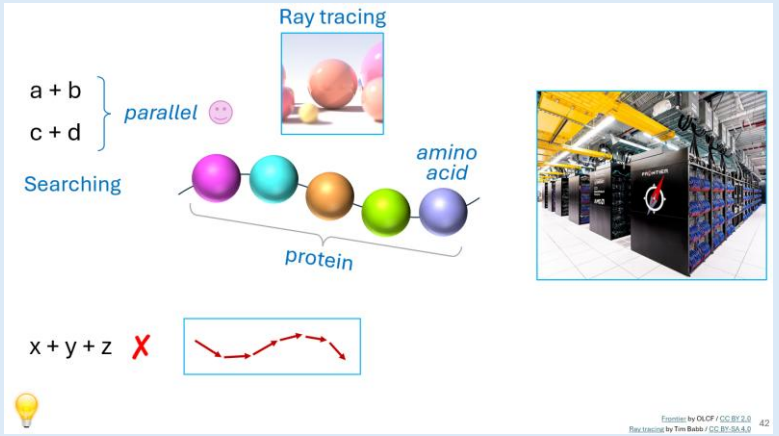

Carbon
Helium
proton +
electron −
neutron
Si
Si
Si
Ar
Balloons
Nitrogen
Oxygen
Fluorine
Neon
Boron
Carbon
Sports Equipment
Basis of Life's Molecules
Protein
Air
Toothpaste
Advertising Signs
Aluminum
Silicon
Phosphorus
Sulfur
Chlorine
Argon
Airplanes
Stone, Sand and Soil
Bones
Eggs
Swimming Pools
Light Bulbs
Gallium
Germanium
Arsenic
Selenium
Bromine
Krypton
Light-Emitting Diodes (LEDs)
Semiconductor Electronics
Poison
Copiers
Photography Film
Flashlights


Transistor
wear and tear
on / off
+
+


Jan Czochralski
1915
sand
silicon
99.9999999%
quartz
1956
Czochralski process
crystal
Silicon ingot


1959
Robert Noyce
Fairchild
Intel
Photolithography
Etching
Chemical vapour deposition


simple
CPU
+ − × ÷
writing
• arbitrary
• complex
decide
?
Memory
ENIAC


Ray tracing
a + b
c + d
parallel
Searching
amino acid
protein
x + y + z ✗